



COEP Technological University

(COEP Tech)

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(Formerly College of Engineering Pune)

YEAR : SECOND YEAR

SEMISTER: III

SUBJECT: Geo-informatics

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Attribute Data & Tables

Table terminology

The screenshot shows a database table titled "US Counties". The table has columns: OBJECTID, Shape, NAME, STATE_NAM, STATE_FIPS, CNTY_FIP, FIPS, POP2000, and POP. The first 10 rows are visible, showing data for various counties in Minnesota, Washington, Idaho, and Montana. Annotations with red callout boxes explain table terminology:

- Title**: Points to the table name "US Counties".
- Field**: Points to the "NAME" column header.
- Each field is specifically defined and established before any data can be entered.**: A general statement about field definitions.
- Records**: Points to the first 10 rows of data.
- Field definitions control the type of data that can be stored in a field.**: Points to the "Shape" column.

OBJECTID *	Shape *	NAME	STATE_NAM	STATE_FIPS	CNTY_FIP	FIPS	POP2000	POP
1	Polygon	Lake of the Woods	Minnesota	27	077	27077	4522	
2	Polygon	Ferry	Washington	53	019	53019	7260	
3	Polygon	Stevens	Washington	53	065	53065	40066	
4	Polygon	Okanogan	Washington	53	047	53047	39564	
5	Polygon	Pend Oreille	Washington	53	051	53051	11732	
6	Polygon	Boundary	Idaho	16	021	16021	9871	
7	Polygon	Lincoln	Montana	30	053	30053	18837	
8	Polygon	Flathead	Montana	30	029	30029	74471	
9	Polygon	Glacier	Montana	30	035	30035	13247	
10	Polygon	Toole	Montana	30	101	30101	5267	

Types of tables

- Attribute table
 - Stores attributes of map features
 - Associated with a spatial data layer
 - Has special fields for spatial information
- Standalone table
 - Stores any tabular data
 - Not associated with spatial data
 - OID instead of FID

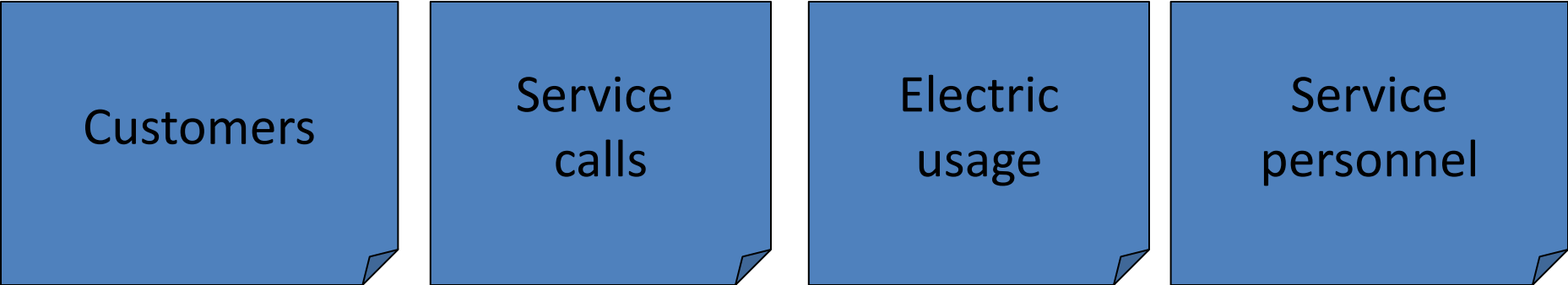
US Counties				
	OBJ	Shape *	NAME	STATE_NAM
▶	1	Polygon	Lake of the Woods	Minnesota
	2	Polygon	Ferry	Washington
	3	Polygon	Stevens	Washington
	4	Polygon	Okanogan	Washington
	5	Polygon	Pend Oreille	Washington
	6	Polygon	Boundary	Idaho

NE_reps					
	OID	OBJECTID	NAME	PARTY	STATE_
	0	169	Chellie Pingree	Democrat	ME
	1	341	James A. Himes	Democrat	CT
	2	343	Christopher S. Murphy	Democrat	CT
	3	370	James P. McGovern	Democrat	MA
	4	371	Joe Courtney	Democrat	CT
	5	372	Patrick J. Kennedy	Democrat	RI
	6	678	James P. McGovern	Democrat	MA

Database Management Systems

Flat file DBMS

- Flat file
 - Stores data as rows of information in files
 - Simple and robust
 - Inefficient for search and query



Customers

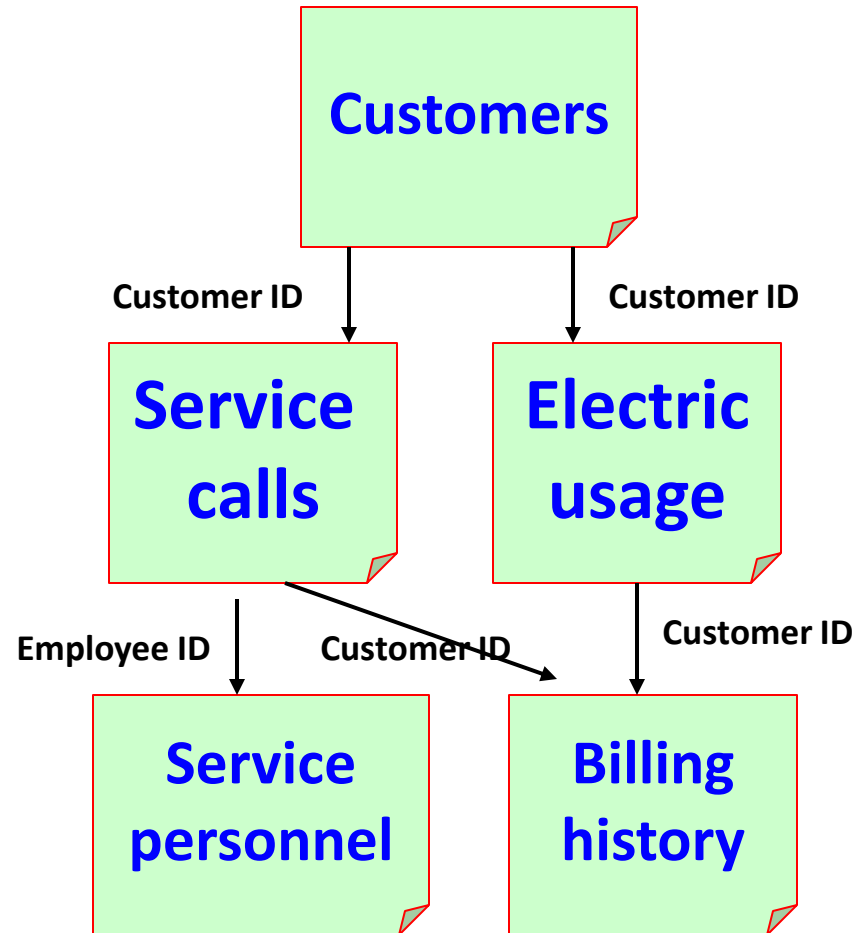
Service
calls

Electric
usage

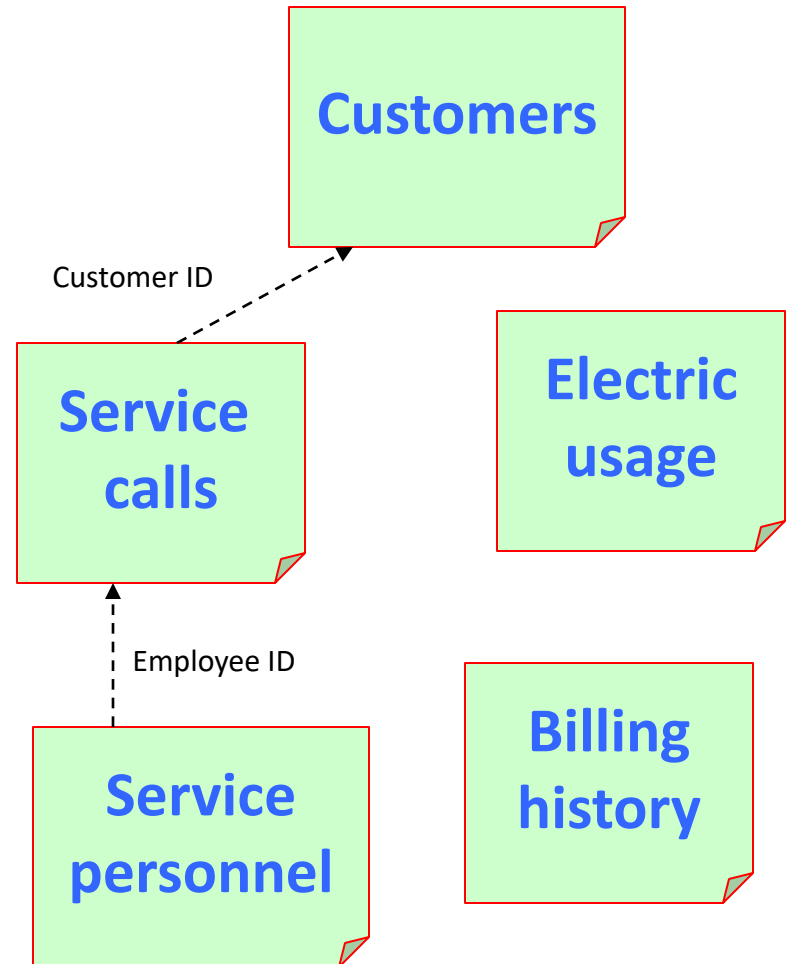
Service
personnel

Hierarchical DBMS

- Stores data in multiple tables
- Tables have defined parent-child relationships
- Pre-set hierarchy of table relationships designed for specific queries
- Very efficient for specific queries
- Range of queries limited by structure



Relational DBMS



Queries- What is a query?

- A query extracts certain records from a table based on a specified condition
- For example,
 - How many students have $\text{GPA} > 3.5$?
 - How many states have population > 1 million?
 - How many counties had greater population in 1990 than in 2000?
 - How many customers have last names beginning with “Mac” or “Mc”?

Structured Query Language, or SQL

- Many databases use a special query language called Structured Query Language
- Can write queries that work in multiple DBMS environments
- Queries can be saved and reused
- Nearly always case-sensitive

SQL Query Examples

Select by Attributes

Enter a WHERE clause to select records in the table window.

Method : Create a new selection

[CNTY_FIPS]
[FIPS]
[POP2000]
[POP00_SQMI]
[POP2010]

= <> Like
> >= And
< <= Or
? * () Not

Is Get Unique Values Go To:

SELECT * FROM counties WHERE:
[POP2010] > 500000

Clear Verify Help Load... Save...
Apply Close

Some Valid Queries

SELECT *FROM cities WHERE
"POP1990" >= 500000

SELECT *FROM counties WHERE
"BEEFCOW_92" < "BEEFCOW_87"

SELECT *FROM parcels WHERE
"LU-CODE" = 42 AND "VALUE" > 50000

SELECT *FROM rentals WHERE
"RENT" > 700 AND "RENT" < 1500

In most databases, SQL expressions are
case-sensitive "Smith" ≠ "SMITH"

Multiple Criteria Queries

- Multiple criteria queries using AND or OR
“STATE” = ‘Alabama’ OR “STATE” = ‘Texas’
[Value] > 5000 AND [Value] < 10000

– Note that the field name must be repeated for each condition

Searching for partial matches

- Sometimes you need to find one string within another rather than an exact match
 - Find all customer names beginning with “Mac” or “Mc”
 - Find all zip codes beginning with 0
- Typically uses a “wildcard” character “*”
 - *Mac* or *Mc*
 - 0*

The Like Operator

- “NAME” LIKE ‘%(D)%’
 - Finds all of the (D) Democrats
- % is the “IGNORE” wildcard
- Ignores Don or Danforth
- “NAME” LIKE ‘%New %’
 - Would find New Hampshire and New York, but not Newcastle or Kennewick

NAME
Sonny Callahan (R)
Terry Everett (R)
Bob Riley (R)
Robert B. Aderholt (R)
Robert E. "Bud" Cramer, Jr. (D)
Spencer Bachus (R)
Earl F. Hilliard (D)
Don Young (R)
Matt Salmon (R)
Ed Pastor (D)
Bob Stump (R)
John Shadegg (R)
Jim Kolbe (R)

NAME
Mesa
Sun City West
Cottonwood-Verde Village
New River
Prescott
Prescott Valley
Flagstaff
Sedona
Payson
Cedar City
Green River
Rock Springs
Magna

Query results

- The result of a query is a set of selected records that meet the criteria
- Subsequent operations on the table will use only the selected records and ignore the others. A user might...
 - Export the selected records to a new table
 - Calculate statistics on the selected records
 - Calculate new values for the selected records
 - Produce a report based on the selected records

Joining tables

	OBJ	Shape *	NAME	STATE_NAM	FIPS
	1	Polygon	Lake of the Woods	Minnesota	27077
	2	Polygon	Ferry	Washington	53019
	3	Polygon	Stevens	Washington	53065
	4	Polygon	Okanogan	Washington	53047
	5	Polygon	Pend Oreille	Washington	53051
	6	Polygon	Boundary	Idaho	16021
	7	Polygon	Lincoln	Montana	30053

	OBJECTID *	FIPS	POP2002	POP2001	POP2000
	1	01001	45604	44698	43903
	2	01003	147932	144787	141410
	3	01005	28826	28993	29047
	4	01007	21838	21935	20869
	5	01009	52968	52143	51213
	6	01011	11367	11454	11613
	7	01013	20911	21157	21336

	OBJ	Shape *	NAME	STATE_NAM	FIPS	OBJE	FIPS	POP2002	POP2001	POP2000
	1	Polygon	Lake of the Woods	Minnesota	27077	1351	27077	4385	4460	4507
	2	Polygon	Ferry	Washington	53019	2962	53019	7268	7290	7292
	3	Polygon	Stevens	Washington	53065	2985	53065	40556	40477	40246
	4	Polygon	Okanogan	Washington	53047	2976	53047	39186	39303	39571
	5	Polygon	Pend Oreille	Washington	53051	2978	53051	12008	11861	11745
	6	Polygon	Boundary	Idaho	16021	560	16021	10085	9946	9924
	7	Polygon	Lincoln	Montana	30053	1623	30053	18555	18691	18840
	8	Polygon	Flathead	Montana	30029	1611	30029	77240	75002	74721

Joined table

Join facts

- Joins are temporary relationships between tables used by a relational DBMS
- Tables must share a common field (key)
- Treats the two tables as a single table
- Original stored data is not affected
- Can be removed when no longer needed

One-to-one joins

Attributes of US States				
	FID	Shape*	STATE_NAME	STATE_ABBR
▶	0	Polygon	Hawaii	HI
	1	Polygon	Washington	WA
	2	Polygon	Montana	MT
	3	Polygon	Maine	ME
	4	Polygon	North Dakota	ND
	5	Polygon	South Dakota	SD
	6	Polygon	Wyoming	WY
	7	Polygon	Wisconsin	WI

quakesum			
STATE	Count	Sum_DAMAGE	Sum_DAMAGE
CA	218	3705234000	
AK	106	32600000	
MT	62	4220000	
WA	67	3775000	
ID	41	1350000	
HI	63	1100000	
OR	24	760000	



Types of Cardinality

Rule of Joining

Attributes of US States				
	FID	Shape*	STATE_NAME	STATE_ABBR
▶	0	Polygon	Hawaii	HI
	1	Polygon	Washington	WA
	2	Polygon	Montana	MT
	3	Polygon	Maine	ME
	4	Polygon	North Dakota	ND
	5	Polygon	South Dakota	SD
	6	Polygon	Wyoming	WY
	7	Polygon	Wisconsin	WI

Destination table

quakesum			
STATE	Count	Sum_DAMAGE	Sum_DEATH
CA	218	3705234000	
AK	106	32600000	
MT	62	4220000	
WA	67	3775000	
ID	41	1350000	
HI	63	1100000	
OR	24	760000	

Source table

US Counties			
Shape*	NAME	STATE_NAME	S
Polygon	Lake of the Woods	Minnesota	27
Polygon	Ferry	Washington	53
Polygon	Stevens	Washington	53
Polygon	Okanogan	Washington	53
Polygon	Pend Oreille	Washington	53
Polygon	Boundary	Idaho	16
Polygon	Lincoln	Montana	30
Polygon	Flathead	Montana	30

Attributes of US States				
	FID	Shape*	STATE_NAME	STATE_ABBR
▶	0	Polygon	Hawaii	HI
	1	Polygon	Washington	WA
	2	Polygon	Montana	MT
	3	Polygon	Maine	ME
	4	Polygon	North Dakota	ND
	5	Polygon	South Dakota	SD
	6	Polygon	Wyoming	WY
	7	Polygon	Wisconsin	WI

One to many

FID	Shape*	STATE_NAME	STATE_ABBR
0	Polygon	Hawaii	HI
1	Polygon	Washington	WA
2	Polygon	Montana	MT
3	Polygon	Maine	ME
4	Polygon	North Dakota	ND
5	Polygon	South Dakota	SD
6	Polygon	Wyoming	WY
7	Polygon	Wisconsin	WI

?

Shape*	NAME	STATE_NAME	S
Polygon	Lake of the Woods	Minnesota	27
Polygon	Ferry	Washington	53
Polygon	Stevens	Washington	53
Polygon	Okanogan	Washington	53
Polygon	Pend Oreille	Washington	53
Polygon	Boundary	Idaho	16
Polygon	Lincoln	Montana	30
Polygon	Flathead	Montana	30

Destination table

Source table

Violates the Rule of Joining

Record to join to destination is ambiguous

Must use a relate instead

No matches?

- Join still occurs, but <Null> values are given to records with no match

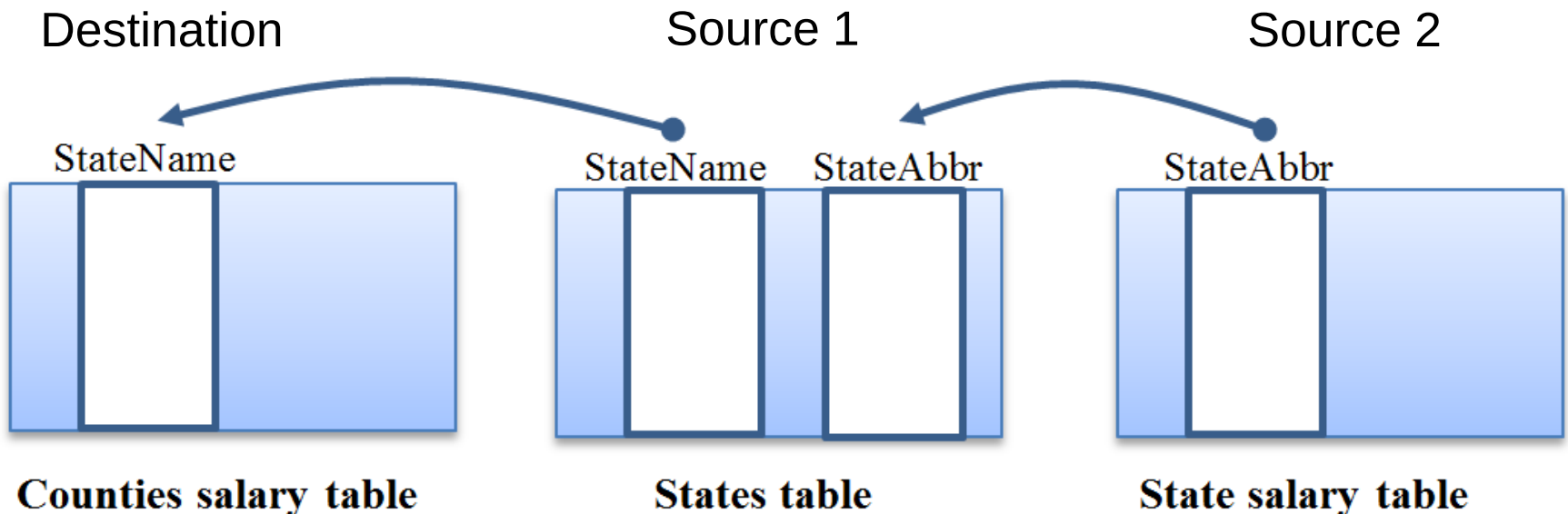
Earthquakes table has
records for Puerto Rico

States table does not have
records for Puerto Rico

quakehis					
	STA	LOCATION	STATE_	STATE	POP201
	PR	Puerto Rico Region	<Null>	<Null>	<Null>
	PR	Puerto Rico Region	<Null>	<Null>	<Null>
	PR	Northwest of Puerto Ri	<Null>	<Null>	<Null>
	AL	Irondale, Alabama	Alabama	AL	473559
	AL	Near Birmingham, Alab	Alabama	AL	473559
	AL	Near Birmingham, Alab	Alabama	AL	473559

Multiple joins

- You can add multiple joins to a destination table using different keys
- You cannot join to a joined source table



Relates

- Similar to a join except that
 - The tables remain separate
 - Items selected in one table may be highlighted in the related table

Related tables

Select the New England states

Table - cd110

US States

NAME	ST	SUB_REGION	ABBR
Hawaii	15	Pacific	HI
Washington	53	Pacific	WA
Montana	30	Mountain	MT
Maine	23	New England	ME
North Dakota	38	West North Central	ND
South Dakota	46	West North Central	SD
Wyoming	56	Mountain	WY
Wisconsin	55	East North Central	WI
Idaho	16	Mountain	ID
Vermont	50	New England	VT
Minnesota	27	West North Central	MN
Oregon	41	Pacific	OR
New Hampshire	33	New England	NH
Iowa	19	West North Central	IA
Massachusetts	25	New England	MA

cd110

NAME	PARTY	DIST	STFI	STA	SQMI	SHA
John F. Tierney	Democrat	2506	25	MA	510.03	25
Edward J. Markey	Democrat	2507	25	MA	178.84	19
Michael E. Capuano	Democrat	2508	25	MA	47.75	13
Niki Tsongas	Democrat	2505	25	MA	582.12	25
Paul W. Hodes	Democrat	3302	33	NH	6662.25	107
Peter Welch	Democrat	5000	50	VT	9613.83	88
Carol Shea-Porter	Democrat	3301	33	NH	2620.1	6
John W. Oliver	Democrat	2501	25	MA	3191.88	59
Richard E. Neal	Democrat	2502	25	MA	951.63	38
Bart Stupak	Democrat	2601	26	MI	25762.8	313
Steve Kagen	Democrat	5508	55	WI	10129.47	126
Michael H. Michaud	Democrat	2302	23	ME	28895.01	226
Don Young	Republican	0200	02	AK	587570.09	2686
Louie Gohmert	Republican	4801	48	TX	8915.78	104
Ted Poe	Republican	4802	48	TX	2054.59	75

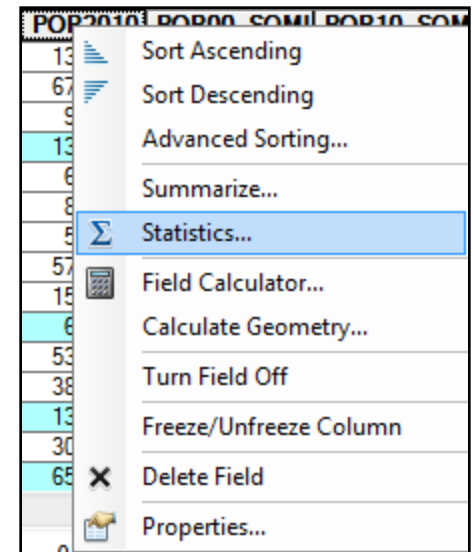
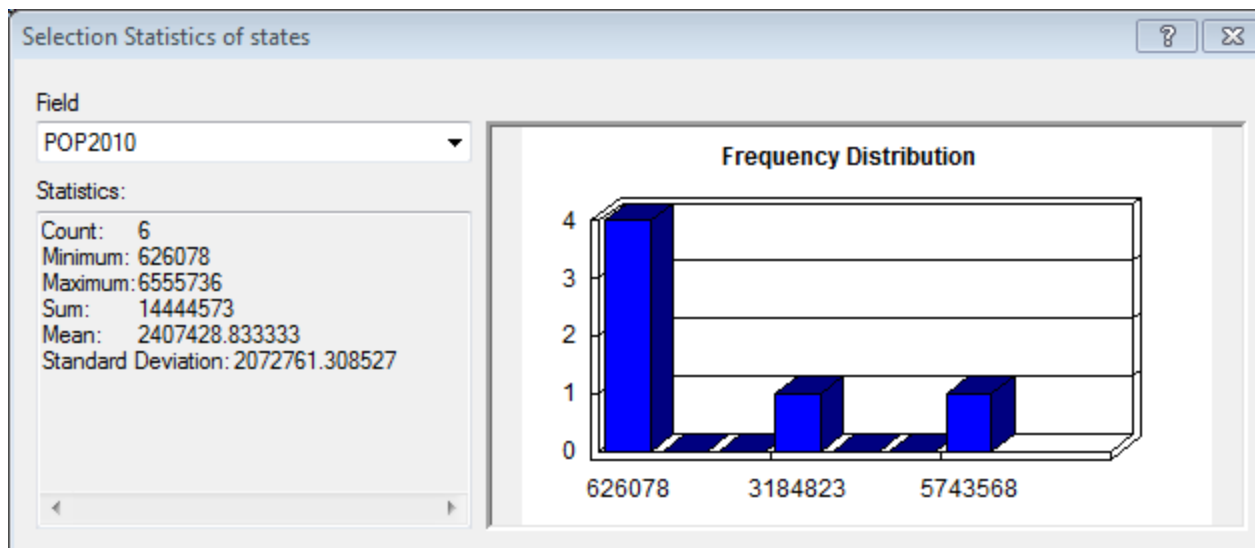
US Counties US States

cd110

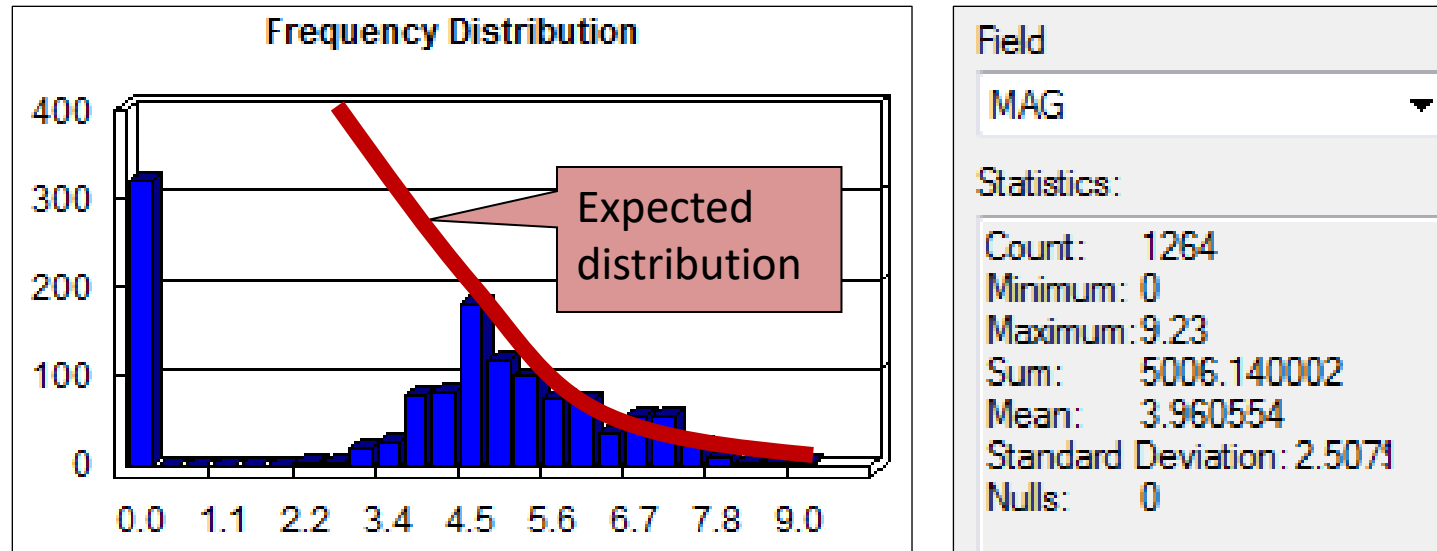
Find reps of New England states

Table statistics

- Often a first step in data analysis
- Opportunity to think critically about data
- Returns statistics only for selected set



Thinking critically



- Why are there so many zero magnitude quakes?
- Why so few small earthquakes?
- What are the implications for data analysis?

Summarizing tables

- Calculate statistics for **groups** of features in a table
- Groups by unique values in the one field
- User chooses statistics to calculate for other fields
- Produces another table as output with groups and stats

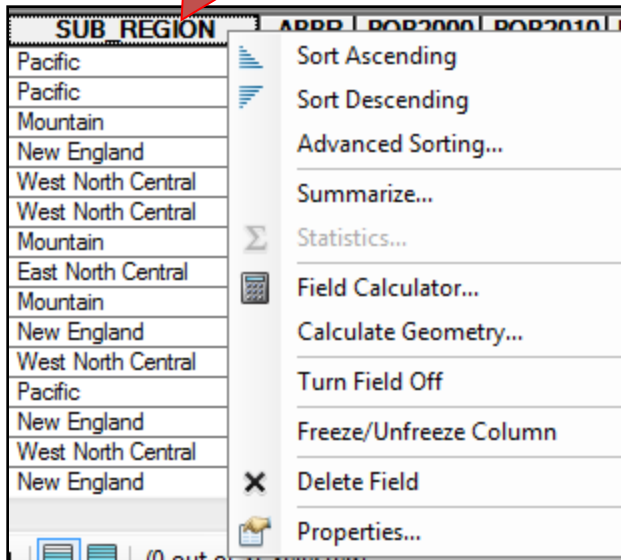
US States						
	NAME	ST	SUB_REGION	ABBR	POP2000	POP2010
	Hawaii	15	Pacific	HI	1211537	1309580
	Washington	53	Pacific	WA	5894121	6756150
	Montana	30	Mountain	MT	902195	983932
	Maine	23	New England	ME	1274923	1338645
	North Dakota	38	West North Central	ND	642200	662194
	South Dakota	46	West North Central	SD	754844	827263
	Wyoming	56	Mountain	WY	493782	548154
	Wisconsin	55	East North Central	WI	5363675	5741617
	Idaho	16	Mountain	ID	1293953	1581697
	Vermont	50	New England	VT	608827	626078
	Minnesota	27	West North Central	MN	4919479	5334772
	Oregon	41	Pacific	OR	3421399	3865839
	New Hampshire	33	New England	NH	1235786	1329915
	Iowa	19	West North Central	IA	2926324	3057995

How many people live in each subregion?

What is the total area of each subregion?

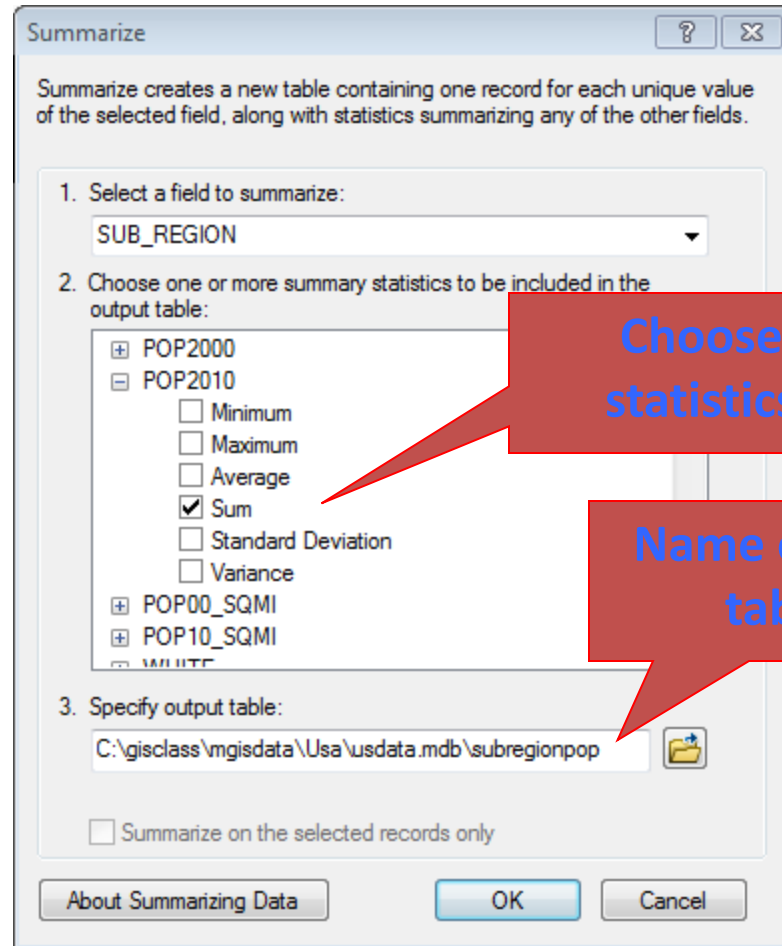
How to summarize

Right-click
Subregion field



Sum Pop2010

Sum SQMI



Choose
statistics

Name output
table

Summarize Output Table

subregionpop				
	OBJECTID *	SUB_REGION	Count_SUB_REGION	Sum_POP2010
▶	1	East North Central	5	47020813
	2	East South Central	4	18438179
	3	Middle Atlantic	3	40940511
	4	Mountain	8	22621196
	5	New England	6	14444573
	6	Pacific	5	50611268
	7	South Atlantic	9	60166524
	8	West North Central	7	20549764
	9	West South Central	4	36420035

Count field always generated
automatically

Create map

- Could we now create a map of population in subregions? No, there are no features.

subregionpop				
	OBJECTID *	SUB_REGION	Count_SUB_REGION	Sum_POP2010
▶	1	East North Central	5	47020813
	2	East South Central	4	18438179
	3	Middle Atlantic	3	40940511
	4	Mountain	8	22621196
	5	New England	6	14444573
	6	Pacific	5	50611268
	7	South Atlantic	9	60166524
	8	West North Central	7	20549764
	9	West South Central	4	36420035

Standalone table

Joining the table

States layer

US States				
	OBJ	Shape *	NAME	SUB_REGION
	1	Polygon	Hawaii	15 Pacific
	2	Polygon	Washington	53 Pacific
	3	Polygon	Montana	30 Mountain
	4	Polygon	Maine	23 New England
	5	Polygon	North Dakota	38 West North Central
	6	Polygon	South Dakota	46 West North Central
	7	Polygon	Wyoming	56 Mountain
	8	Polygon	Wisconsin	55 East North Central
	9	Polygon	Idaho	16 Mountain
	10	Polygon	Vermont	50 New England
	11	Polygon	Minnesota	27 West North Central
	12	Polygon	Oregon	41 Pacific
	13	Polygon	New Hampshire	33 New England

Summarized table

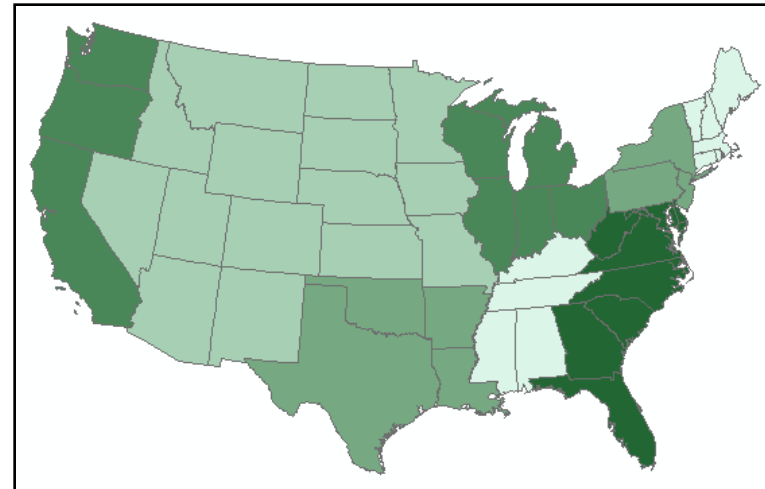
subregionpop				
	OBJECTID *	SUB_REGION	Count_SUB_REGION	Sum_POP2010
	1	East North Central	5	47020813
	2	East South Central	4	18438179
	3	Middle Atlantic	3	40940511
	4	Mountain	8	22621196
	5	New England	6	14444573
	6	Pacific	5	50611268
	7	South Atlantic	9	60166524
	8	West North Central	7	20549764
	9	West South Central	4	36420035

Fields

Value:

Normalization:

Color Ramp:



Database storage concepts

Fields

- Fields have specific types available
- Must be defined before use
- Once defined, cannot be changed
- Naming rules
 - No more than 13 characters
 - Use only letters and numbers
 - Must start with a letter

ArcGIS field data types

Field Type	Explanation	Examples
Short	Integers stored as 2-byte binary numbers <i>Range of values -32,000 to +32,000</i>	255 12001
Long	Integers stored as 10-byte binary numbers <i>Range of values -2.14 billion to +2.14 billion</i>	156000 457890
Float	Floating-point values with eight significant digits in the mantissa	1.289385e12 1.5647894e-02
Double	Double-precision floating-point values with 16 significant digits in the mantissa	1.12114118119141e13
Text	Alphanumeric strings	'Maple St' 'John H. Smith'
Date	Date format	07/12/92 10/17/63
BLOB*	Binary large object; any complex binary data including images, documents, etc.	

Time to think

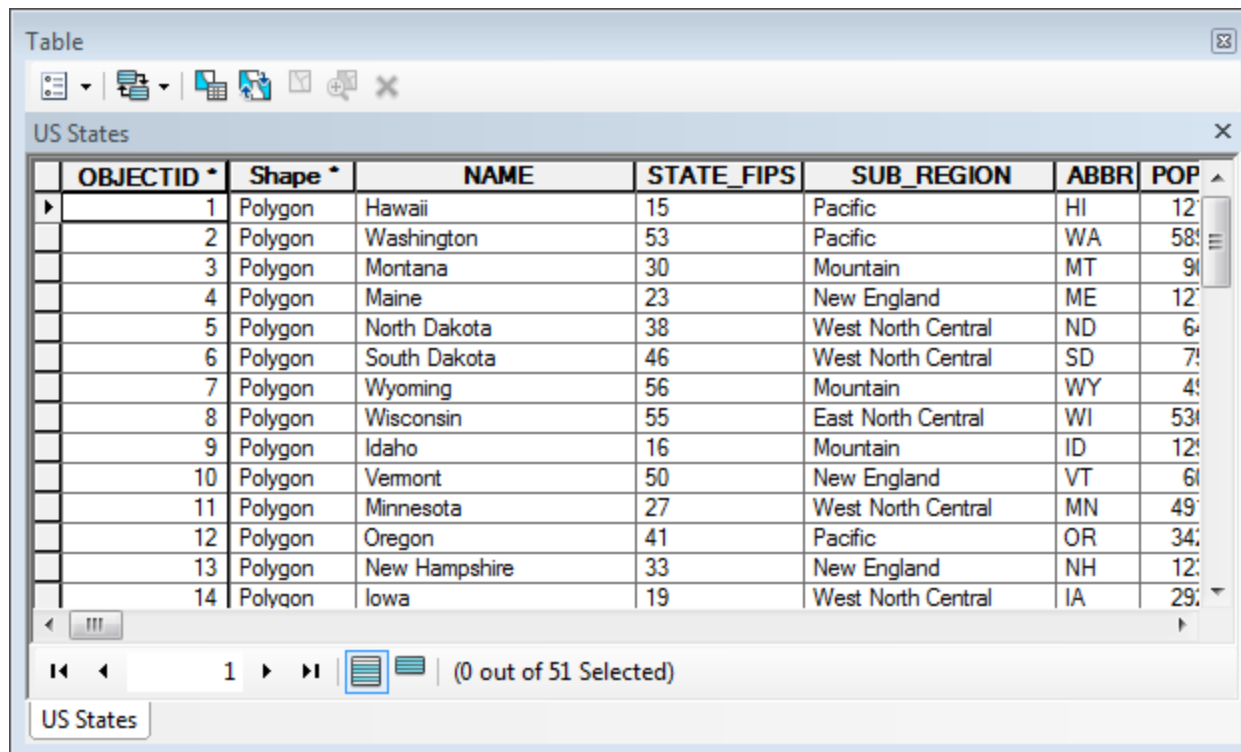
- What kind of storage would be used to store the following kinds of data? How many bytes are needed for each?

- Last name
- Number of children in household
- Student grade point average
- Year of birth
- County population
- Aerial image of county

Field Type
Short
Long
Float
Double
Text
Date
BLOB*

Tables in ArcGIS

- Tables contain attribute data
- Many formats, one interface



The screenshot shows the 'Table' window in ArcGIS, displaying a table titled 'US States'. The table has 8 columns: OBJECTID, Shape, NAME, STATE_FIPS, SUB_REGION, ABBR, and POP. The data is sorted by OBJECTID. The table is currently showing rows 1 through 14, with a scrollbar indicating more rows are available. The status bar at the bottom shows '(0 out of 51 Selected)'.

OBJECTID	Shape	NAME	STATE_FIPS	SUB_REGION	ABBR	POP
1	Polygon	Hawaii	15	Pacific	HI	12
2	Polygon	Washington	53	Pacific	WA	58
3	Polygon	Montana	30	Mountain	MT	90
4	Polygon	Maine	23	New England	ME	12
5	Polygon	North Dakota	38	West North Central	ND	6
6	Polygon	South Dakota	46	West North Central	SD	7
7	Polygon	Wyoming	56	Mountain	WY	4
8	Polygon	Wisconsin	55	East North Central	WI	53
9	Polygon	Idaho	16	Mountain	ID	12
10	Polygon	Vermont	50	New England	VT	6
11	Polygon	Minnesota	27	West North Central	MN	49
12	Polygon	Oregon	41	Pacific	OR	34
13	Polygon	New Hampshire	33	New England	NH	12
14	Polygon	Iowa	19	West North Central	IA	29

Sources of tables

- Dbase files
- INFO files
- ASCII Text files (tab or comma delimited)
- Records from SQL database systems
- Excel worksheets

ArcMap table interface

Options menu

Field

Right-click field name to get menu

Title

Records

Status bar

The screenshot displays the ArcMap Table interface. The table is titled 'States' and contains three columns: 'STATE_NAME', 'STATE_FIPS', and 'SUB_REGION'. The data rows list various US states and their corresponding FIPS codes and sub-regions. A right-click context menu is open over the 'SUB_REGION' field, showing options such as 'Sort Ascending', 'Sort Descending', 'Advanced Sorting...', 'Summarize...', 'Statistics...', 'Field Calculator...', 'Calculate Geometry...', 'Turn Field Off', 'Freeze/Unfreeze Column', 'Delete Field', and 'Properties...'. The status bar at the bottom indicates '(0 out of 51 Selected)'.

STATE_NAME	STATE_FIPS	SUB_REGION
Hawaii	15	Pacific
Washington	53	Pacific
Montana	30	Mountain
Maine	23	New England
North Dakota	38	West North Central
South Dakota	46	West North Central
Wyoming	56	Mountain
Wisconsin	55	East North Central
Idaho	16	Mountain

Options menu

Field

Right-click field name to get menu

Title

Records

Status bar

Sort Ascending

Sort Descending

Advanced Sorting...

Summarize...

Statistics...

Field Calculator...

Calculate Geometry...

Turn Field Off

Freeze/Unfreeze Column

Delete Field

Properties...

(0 out of 51 Selected)

Managing multiple tables

The screenshot shows a GIS application interface with multiple table windows. The top window, titled 'Table', displays a table named 'cd110' with the following data:

OBJECTID	Shape	NAME	PARTY	DISTRICTID	STFIPS	STATE
1	Polygon	Robert A. Brady	Democrat	4201	42	PA
2	Polygon	Chaka Fattah	Democrat	4202	42	PA
3	Polygon	Phil English	Republican	4203	42	PA
4	Polygon	Jason Altmire	Democrat	4204	42	PA
5	Polygon	John E. Peterson	Republican	4205	42	PA
6	Polygon	Jim Gerlach	Republican	4206	42	PA
7	Polygon	Joe Sestak	Democrat	4207	42	PA
8	Polygon	Patrick J. Murphy	Democrat	4208	42	PA

Below this table, a context menu titled 'Arrange Tables' is open, showing options: 'New Horizontal Tab Group', 'New Vertical Tab Group', 'Move to Previous Tab Group', and 'Move to Next Tab Group'. The 'New Vertical Tab Group' option is highlighted.

At the bottom, two additional table windows are shown side-by-side. The left window, titled 'US Counties', displays a table with the following data:

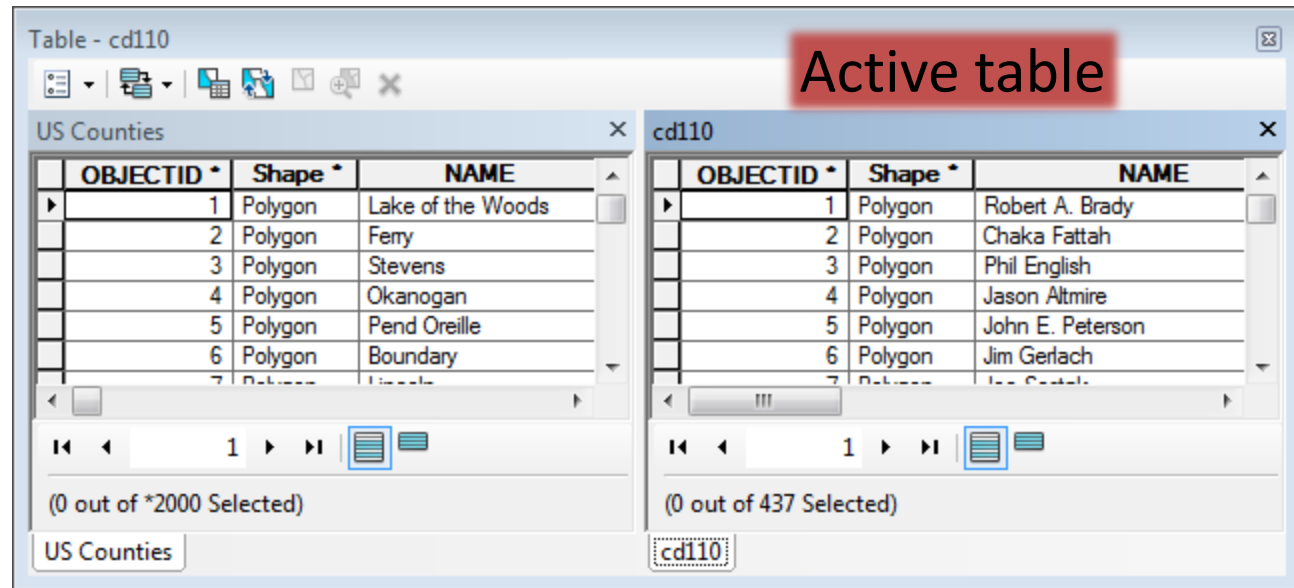
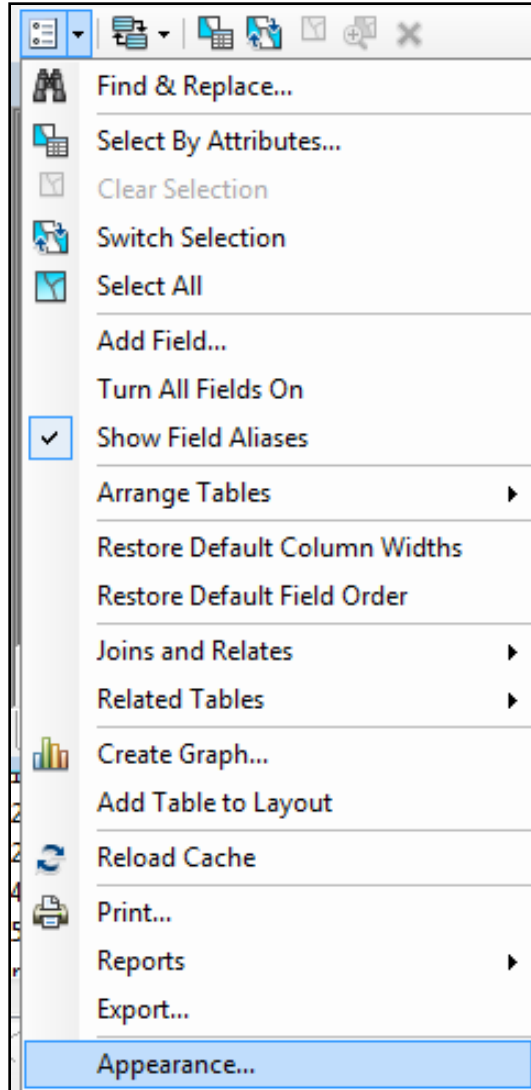
OBJECTID	Shape	NAME
1	Polygon	Lake of the Woods
2	Polygon	Ferry
3	Polygon	Stevens
4	Polygon	Okanogan
5	Polygon	Pend Oreille
6	Polygon	Boundary

The right window, titled 'cd110', displays a table with the following data:

OBJECTID	Shape	NAME
1	Polygon	Robert A. Brady
2	Polygon	Chaka Fattah
3	Polygon	Phil English
4	Polygon	Jason Altmire
5	Polygon	John E. Peterson
6	Polygon	Jim Gerlach

Annotations include a blue circle around the 'cd110' tab in the 'US Counties' window and an arrow pointing to the 'cd110' tab in the 'cd110' window, with the text 'Switch between' below them. The text 'Arrange in tabs' is also present on the right side of the image.

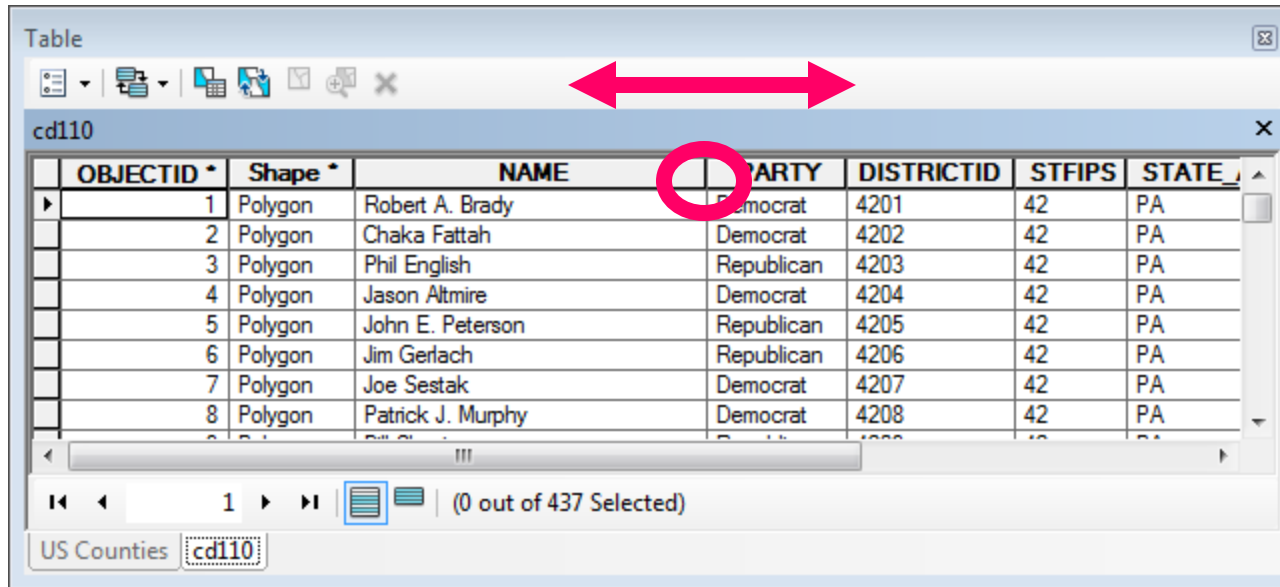
Table Options menu



Commands affect the currently active table

Adjusting field width

- Temporary, does not affect stored file



The screenshot shows a QGIS Table window titled 'Table' with a toolbar. Below the toolbar is a tab labeled 'cd110'. The table contains the following data:

	OBJECTID *	Shape *	NAME	PARTY	DISTRICTID	STFIPS	STATE_
▶	1	Polygon	Robert A. Brady	Democrat	4201	42	PA
	2	Polygon	Chaka Fattah	Democrat	4202	42	PA
	3	Polygon	Phil English	Republican	4203	42	PA
	4	Polygon	Jason Altmire	Democrat	4204	42	PA
	5	Polygon	John E. Peterson	Republican	4205	42	PA
	6	Polygon	Jim Gerlach	Republican	4206	42	PA
	7	Polygon	Joe Sestak	Democrat	4207	42	PA
	8	Polygon	Patrick J. Murphy	Democrat	4208	42	PA

At the bottom of the window, there is a status bar showing 'US Counties' and 'cd110' in a dropdown menu, and '(0 out of 437 Selected)'.

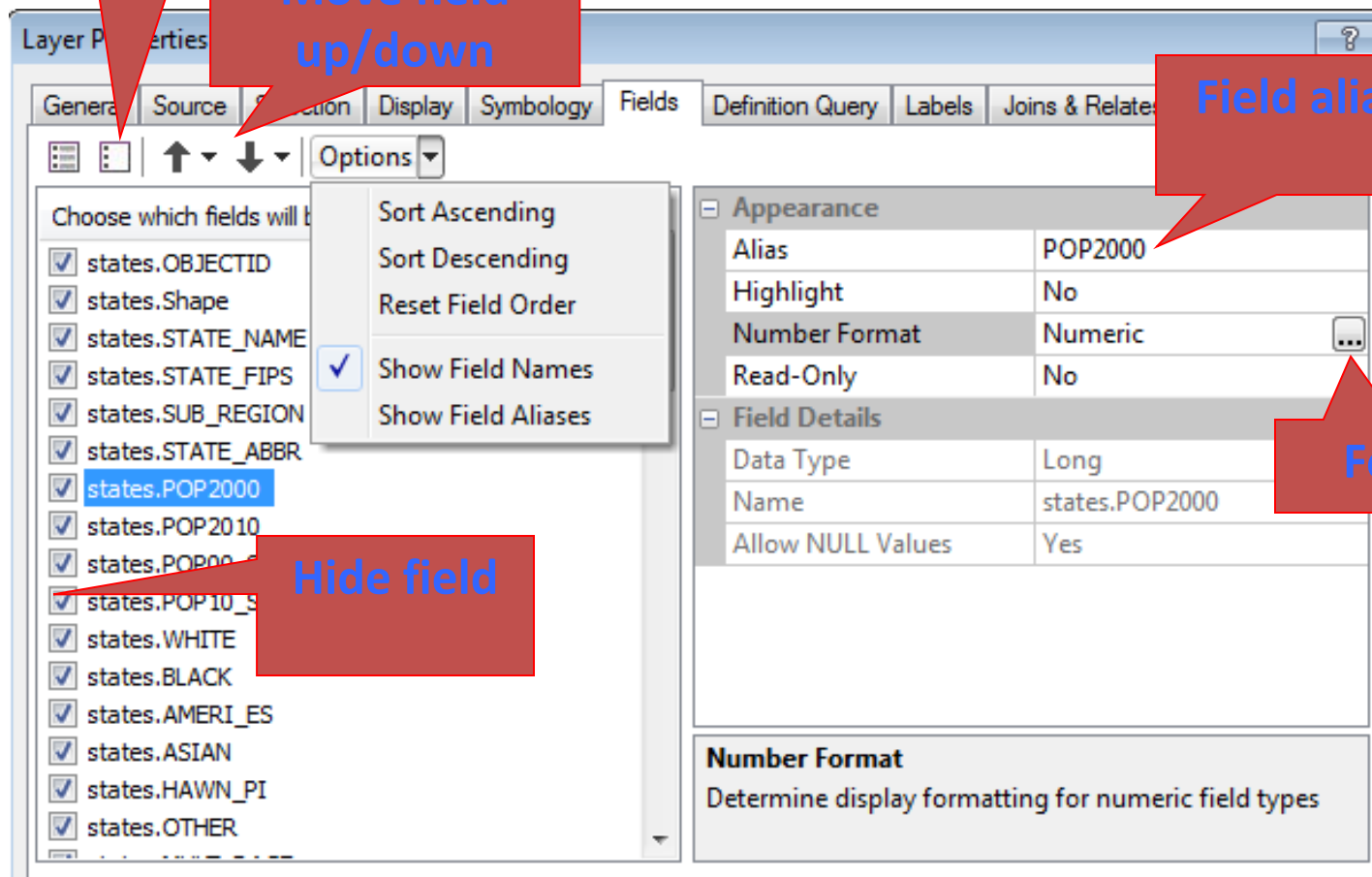
Hover over field break to get double arrow, then drag

Field properties tab

Show/hide
all

Move field
up/down

Field alias



Hide field

Format

Shortcut to field properties

Right-click
field name

US States

	OBJECTID *	Shape *	NAME	STATE FIPS	SUB_REGION
	1	Polygon	Hawaii		
	2	Polygon	Washington		
	3	Polygon	Montana		
	4	Polygon	Maine		
	5	Polygon	North Dakota		
	6	Polygon	South Dakota		
	7	Polygon	Wyoming		
	8	Polygon	Wisconsin		
	9	Polygon	Idaho		
	10	Polygon	Vermont		
	11	Polygon	Minnesota		
	12	Polygon	Oregon		
	13	Polygon	New Hampshire		
	14	Polygon	Iowa		

Right-click context menu options:

- Sort Ascending
- Sort Descending
- Advanced Sorting...
- Summarize...
- Statistics...
- Field Calculator...
- Calculate Geometry...
- Turn Field Off
- Freeze/Unfreeze Column
- Delete Field
- Properties...

Field Properties

Name: states.STATE_NAME

Alias: NAME

Type: String

Display

☐ Turn field off

☐ Make field read only

☐ Highlight field

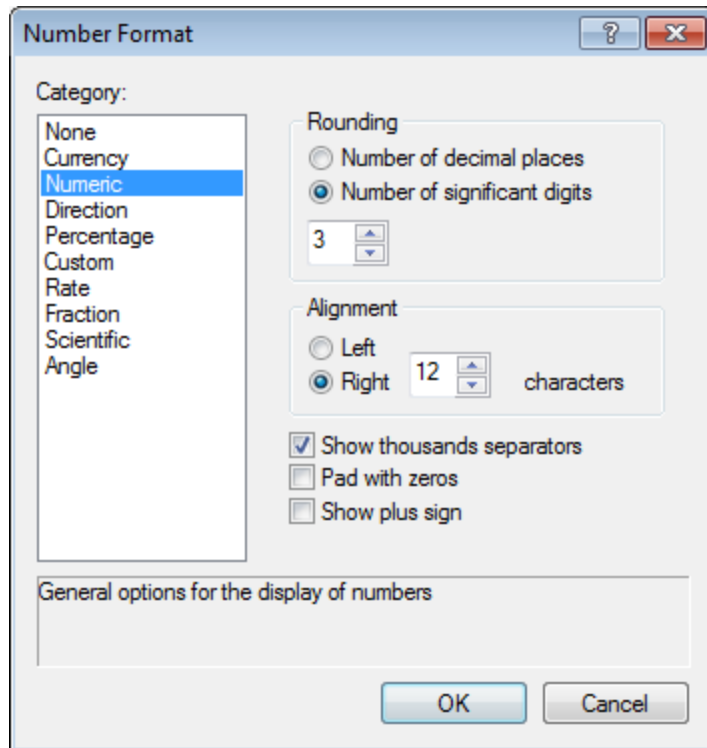
Number Format: ...

Data

Allow NULL Values	Yes
Default Value	
Length	25

OK Cancel Apply

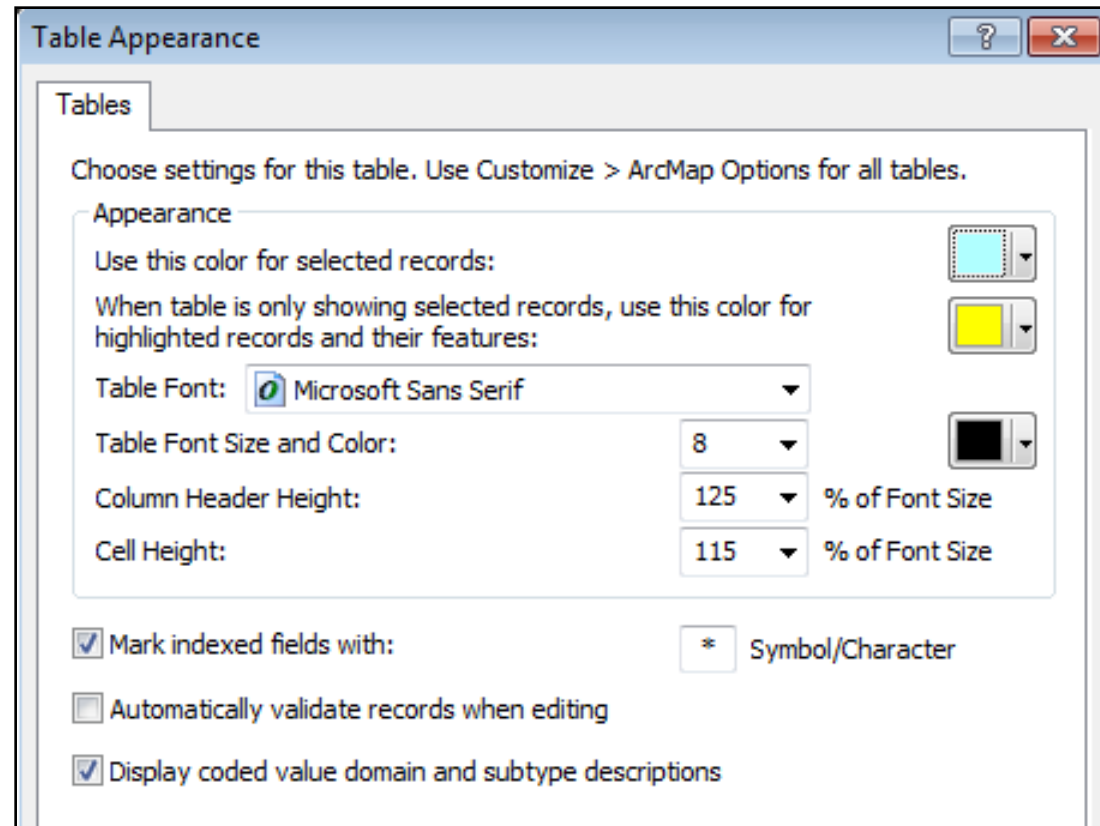
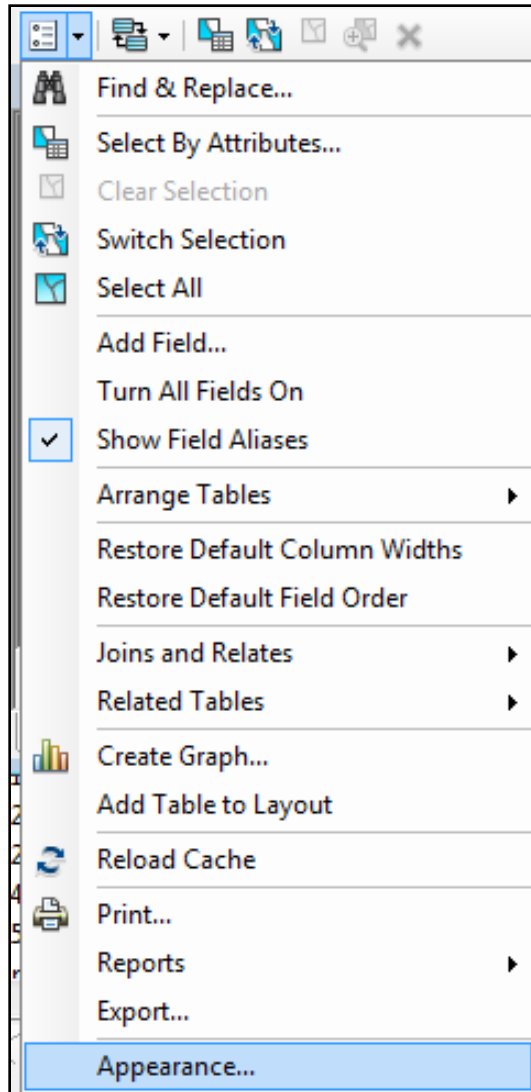
Formatting field display



POP2010
1309580
6756150
983932
1338645
662194
827263
548154

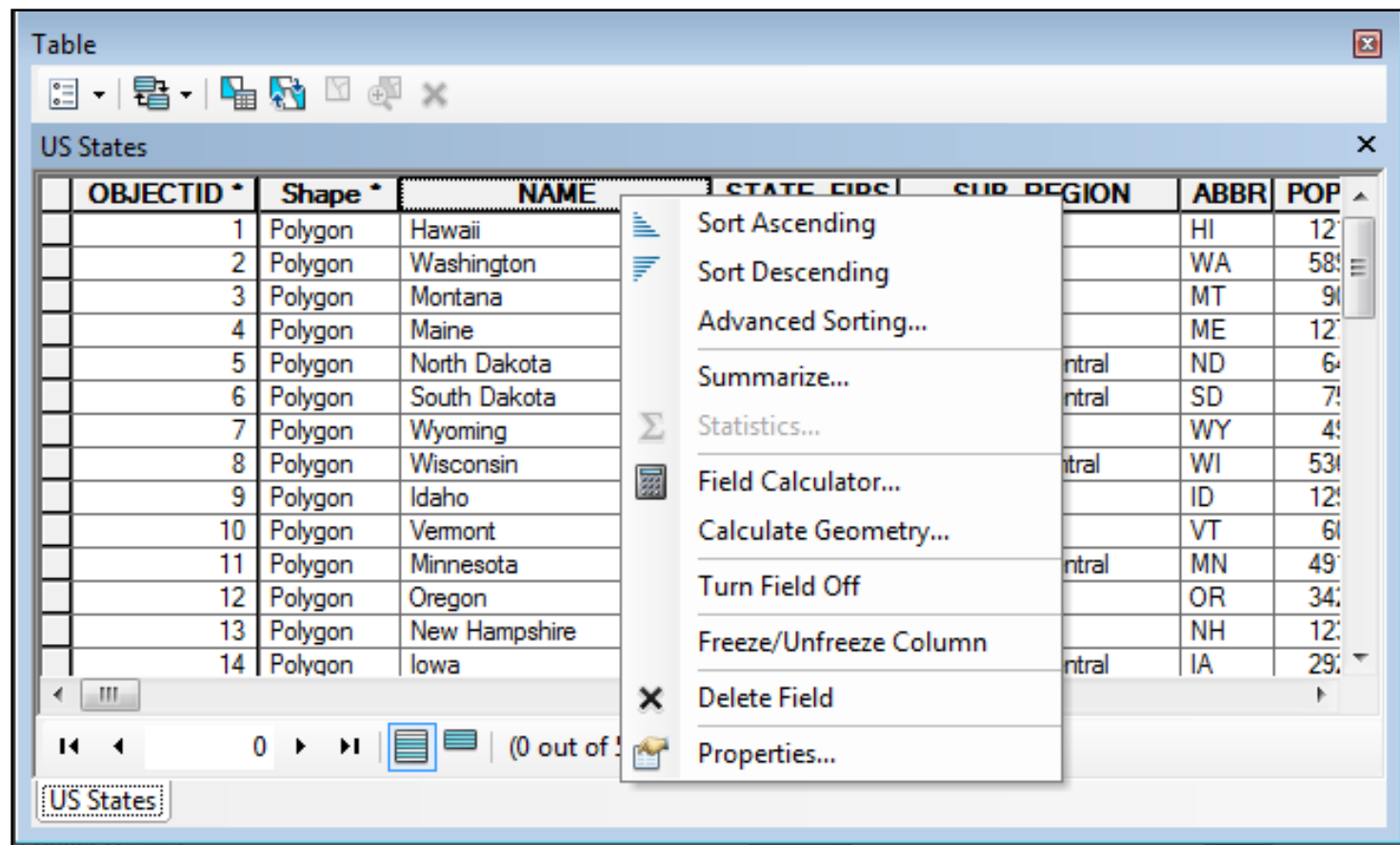
POP2010
1,310,000
6,760,000
984,000
1,340,000
662,000
827,000
548,000

Table appearance

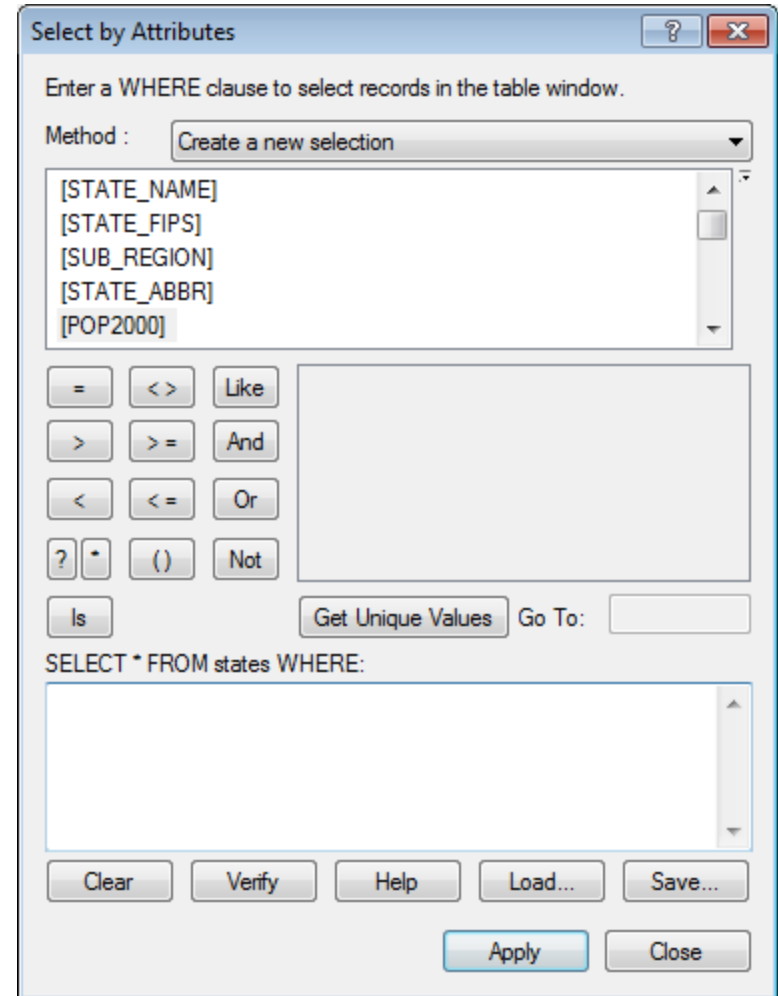
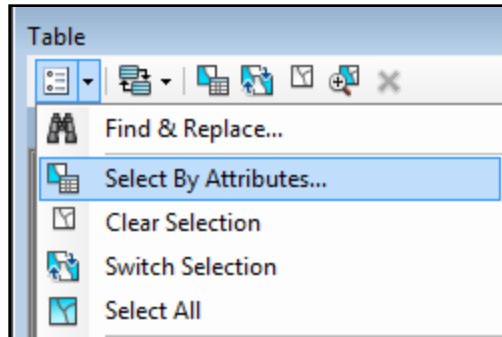


Sorting tables

- Has no effect on original data



Select by Attributes



Some valid queries

[POP1990] > 1000000

"STATE_NAME" = 'Alabama'

[POP2000] >= [POP1990]

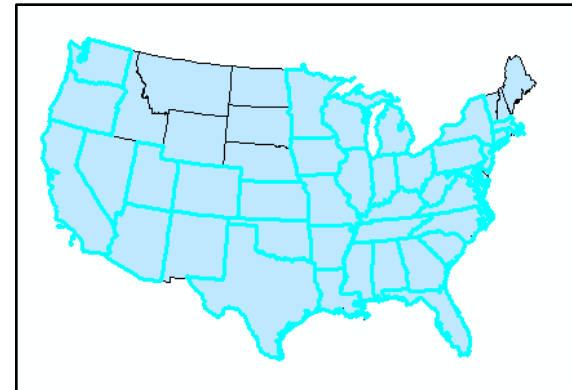
DBF tables have field names
enclosed in quotes

Geodatabase tables have field
names enclosed in brackets

Show selected

```
SELECT * FROM states WHERE:
```

```
"POP2000" > 2000000
```



Table

US States

	OBJ	Shape *	NAME	ST	SUB_REGION	ABBR	POP2000	POP2010
	7	Polygon	Wyoming	56	Mountain	WY	493782	548,000
	8	Polygon	Wisconsin	55	East North Central	WI	5363675	5,740,000
	9	Polygon	Idaho	16	Mountain	ID	1202053	1,500,000
	10	Polygon	Vermont	50	New England	VT	62398	623,000
	11	Polygon	Minnesota	27	West North Central	MN	4919479	5,330,000
	12	Polygon	Oregon	41	Pacific	OR	3421399	3,870,000
	13	Polygon	New Hampshire	33	New England	NH	1897645	1,950,000
	14	Polygon	Iowa	19	West North Central	IA	2926324	3,060,000
	15	Polygon	Massachusetts	25	New England	MA	6349097	6,560,000
	16	Polygon	Nebraska	31	West North Central	NE	1897645	1,950,000

US States

(34 out of 51 Selected)

Table

US States

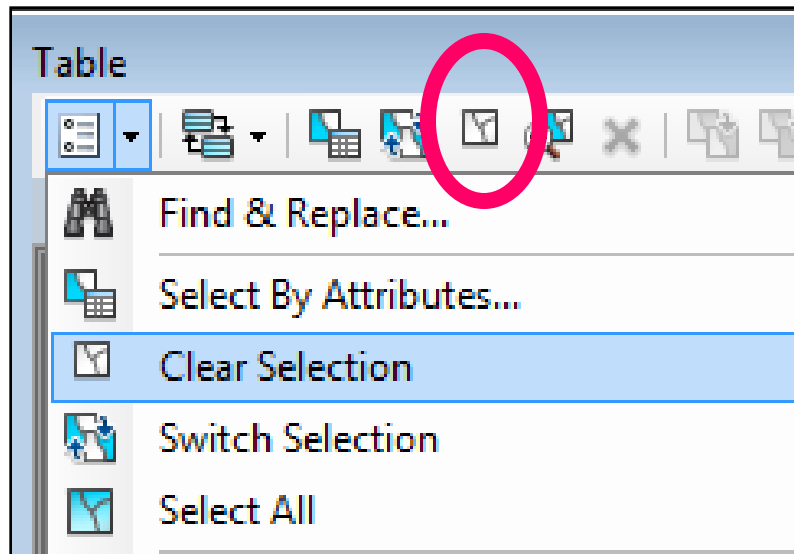
	OBJ	Shape *	NAME	ST	SUB_REGION	ABBR	POP2000	POP2010
	2	Polygon	Washington	53	Pacific	WA	5894121	6,760,000
	8	Polygon	Wisconsin	55	East North Central	WI	5363675	5,740,000
	11	Polygon	Minnesota	27	West North Central	MN	4919479	5,330,000
	12	Polygon	Oregon	41	Pacific	OR	3421399	3,870,000
	14	Polygon	Iowa	19	West North Central	IA	2926324	3,060,000
	15	Polygon	Massachusetts	25	New England	MA	6349097	6,560,000
	17	Polygon	New York	36	Middle Atlantic	NY	18976457	19,500,000
	18	Polygon	Pennsylvania	42	Middle Atlantic	PA	12281054	12,600,000
	19	Polygon	Connecticut	09	New England	CT	3405565	3,540,000
	21	Polygon	New Jersey	34	Middle Atlantic	NJ	8414350	8,820,000

US States

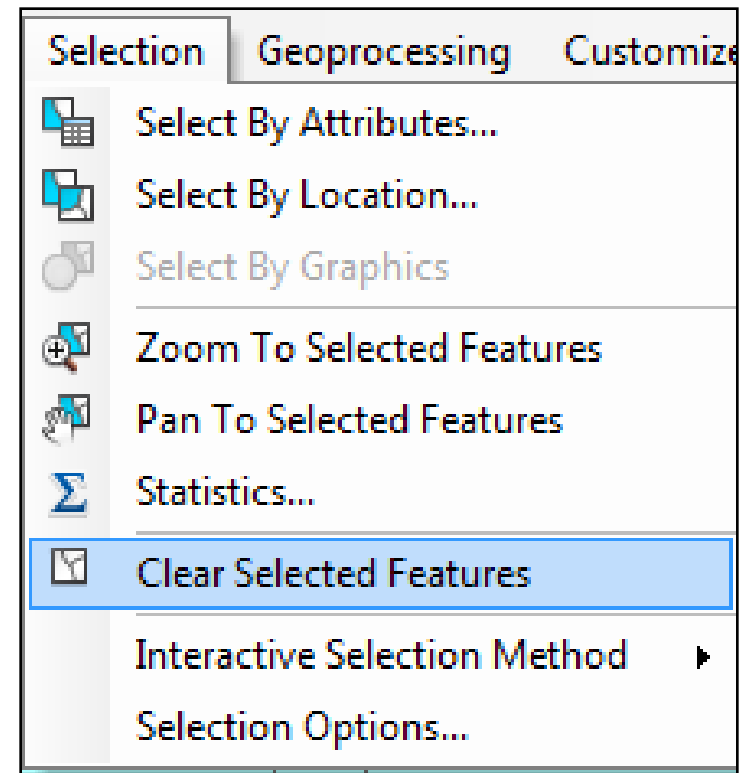
(34 out of 51 Selected)

Clear Selection

In Table window



From main menu

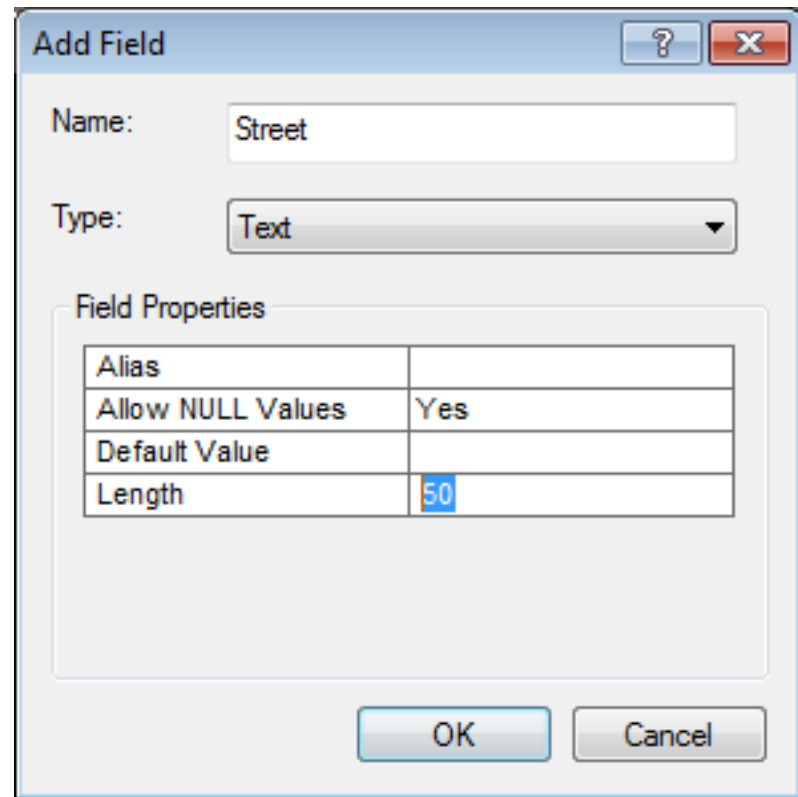
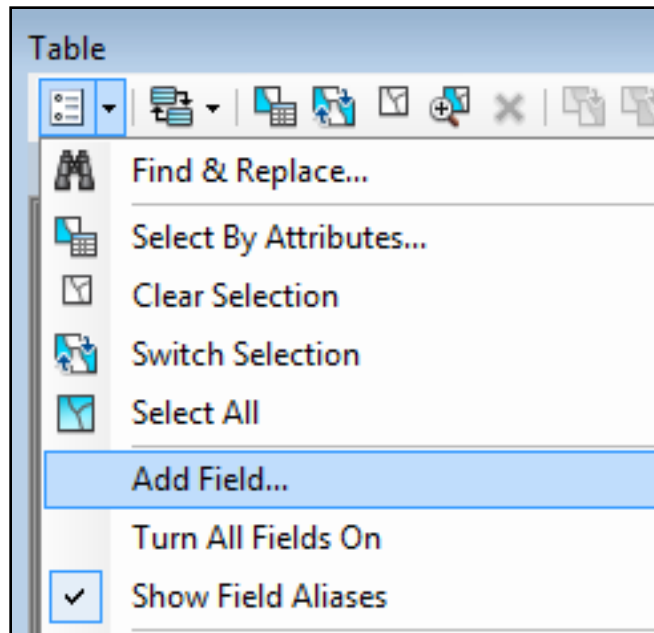


ArcGIS field data types

Field Type	Explanation	Examples
Short	Integers stored as 2-byte binary numbers <i>Range of values -32,000 to +32,000</i>	255 12001
Long	Integers stored as 10-byte binary numbers <i>Range of values -2.14 billion to +2.14 billion</i>	156000 457890
Float	Floating-point values with eight significant digits in the mantissa	1.289385e12 1.5647894e-02
Double	Double-precision floating-point values with 16 significant digits in the mantissa	1.12114118119141e13
Text	Alphanumeric strings	'Maple St' 'John H. Smith'
Date	Date format	07/12/92 10/17/63
BLOB*	Binary large object; any complex binary data including images, documents, etc.	

Adding a field

- In ArcMap

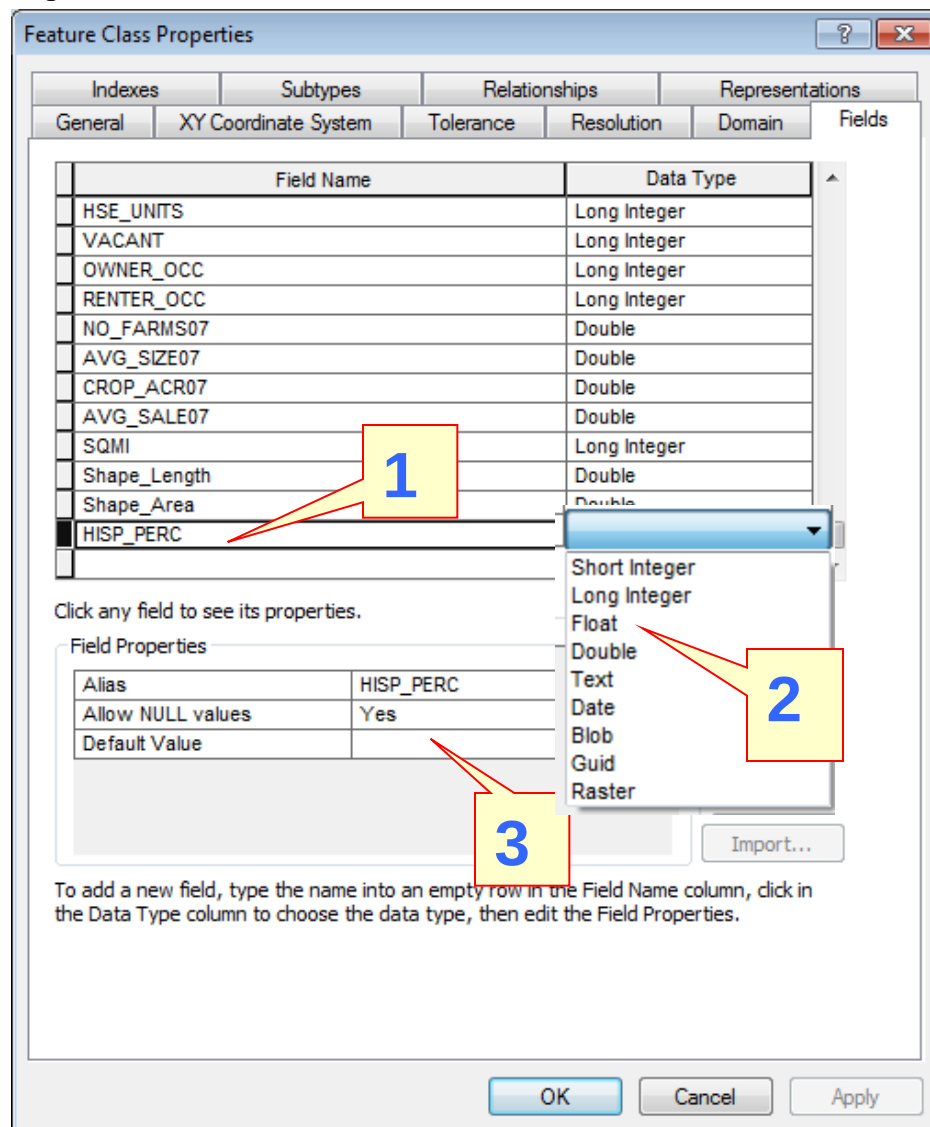


Think about the field type before you add the field!

Adding multiple fields

- Easier in the Catalog properties

- Type name in empty row
- Set field type
- Set field properties
- Click Apply
- Add next...

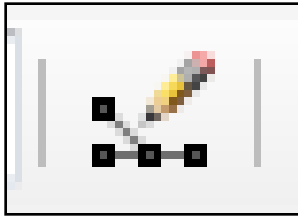


Think about the field type
before you add the field!

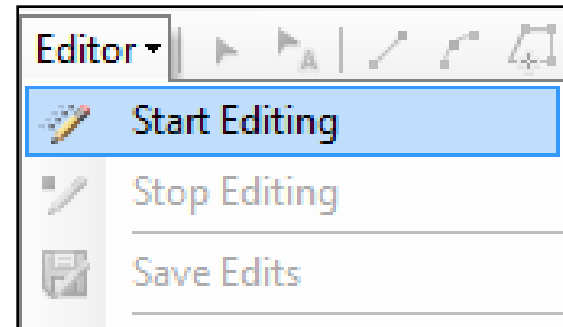
Editing and calculating fields

Editing fields

Open Editor toolbar



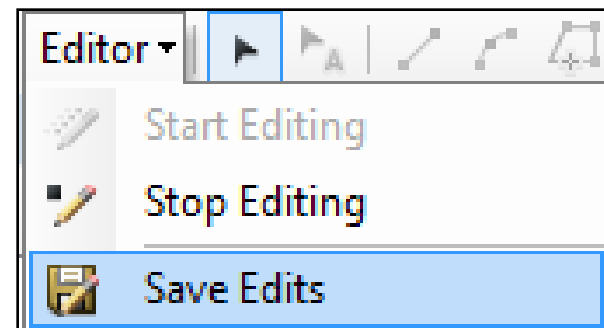
Start editing



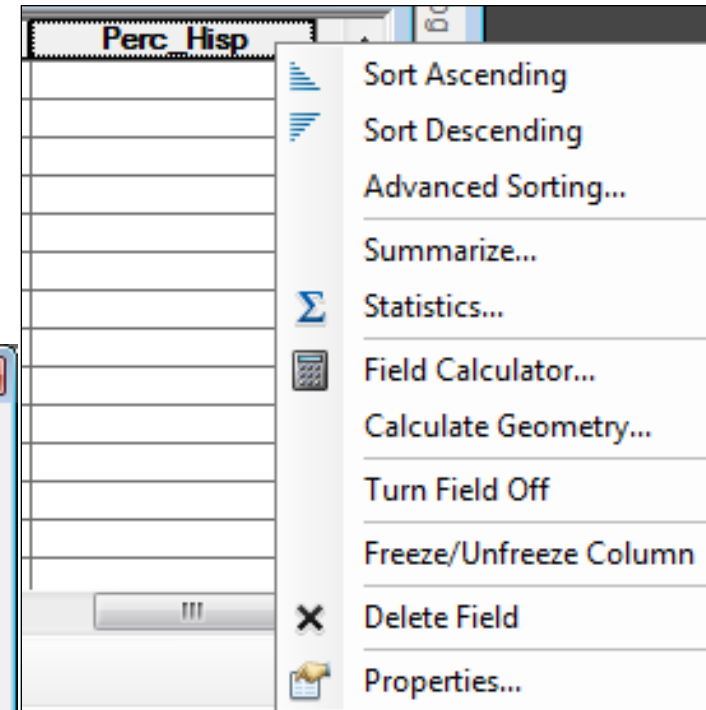
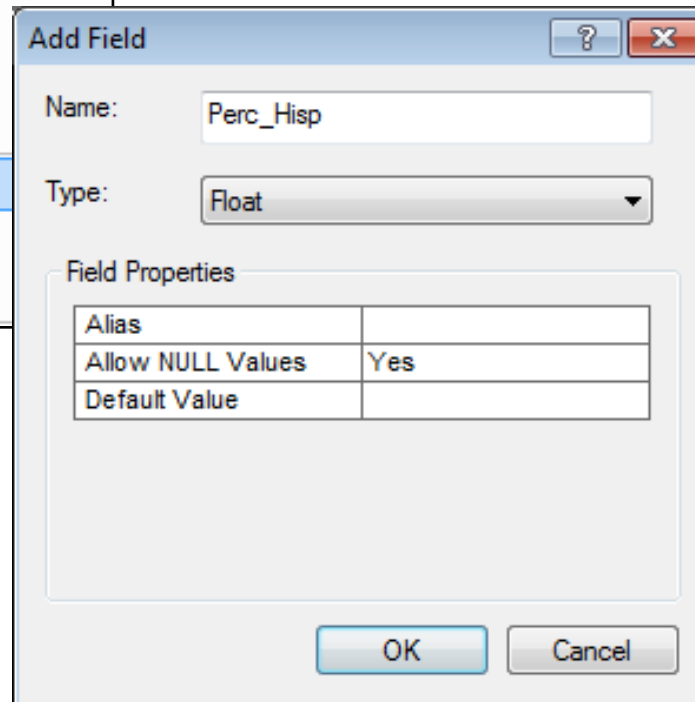
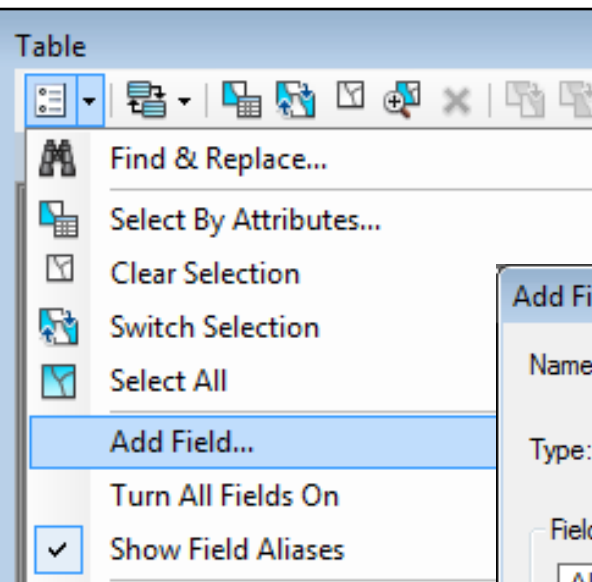
Type edits in fields

	Ave_MAG	Ave_MMI	Risk
5	6.5042	3.7358	High
0	3.6143	6.1429	Low
7	5.0113	6.5625	High
0	2.135	5.9	Lo
0	6.35	7	
0	6.3167	8.5	

Save edits, stop editing

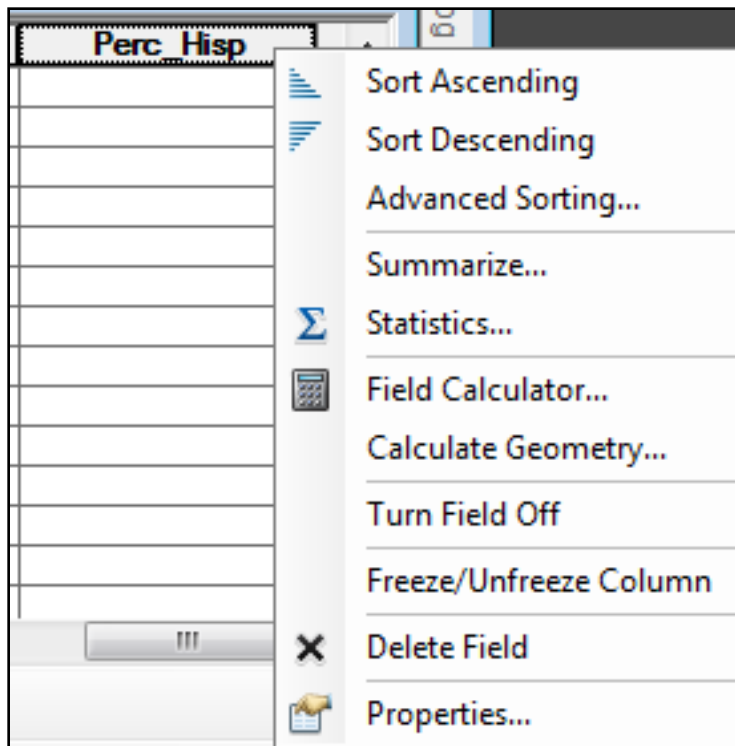


Calculating fields



Add a new field if necessary
Consider whether you need decimal places!

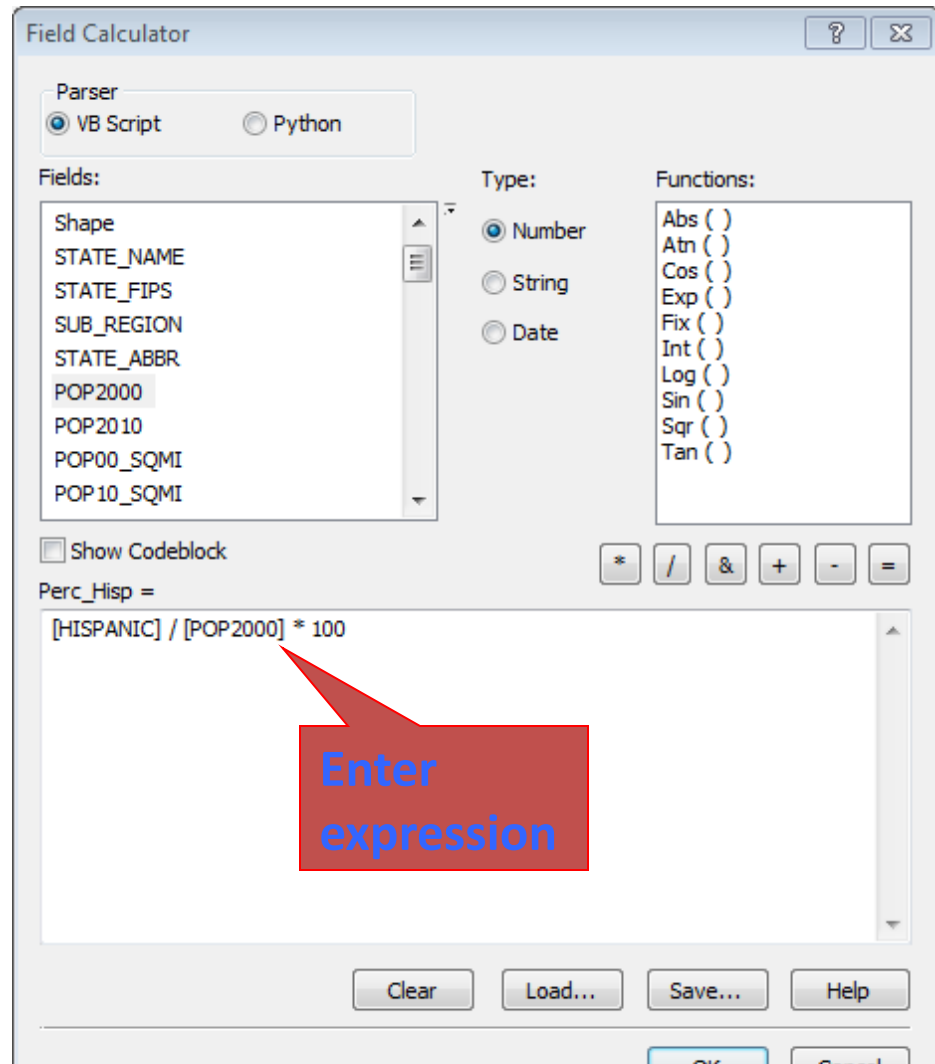
Calculate



Right-click field to calculate

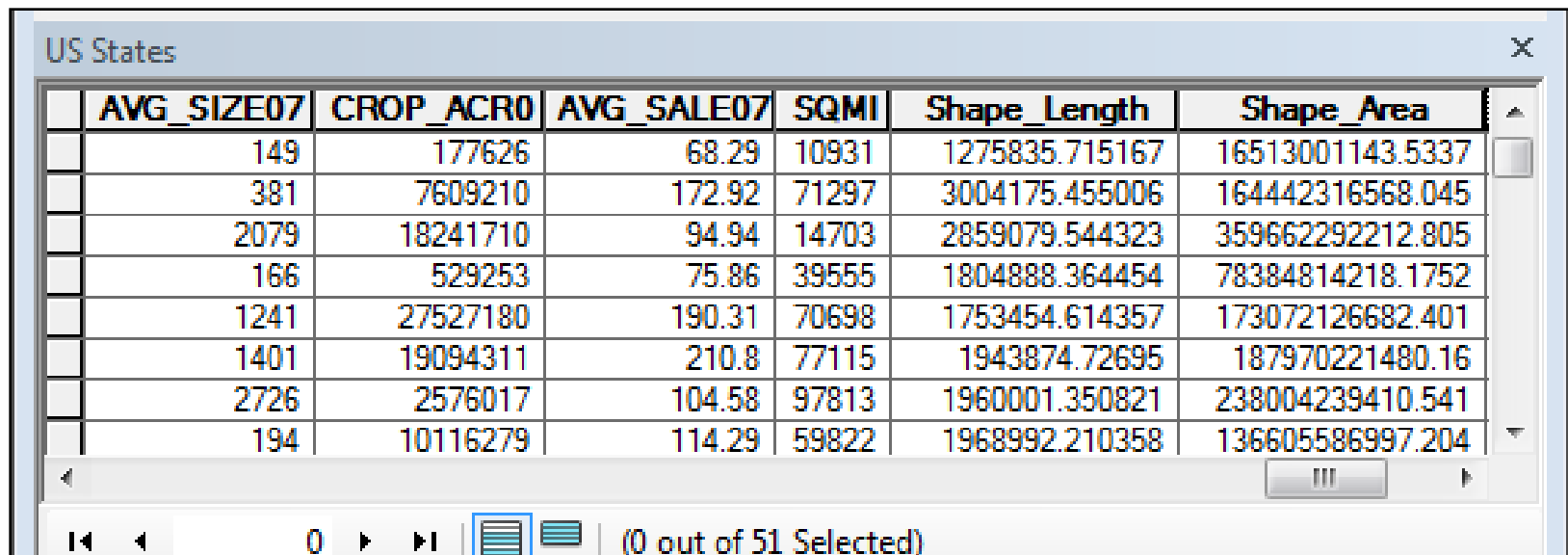
Usually field should be
EMPTY

Be careful! Calculations
can't be undone unless you
are in an edit session



Geodatabase geometry measures

- Automatically created and maintained
 - Usually appear at end of table
 - Shape_Area field
 - Shape_Length field
 - Units will match units of the coordinate system



The screenshot shows a database table window titled "US States". The table contains several columns, with the last two being geometry measures: "Shape_Length" and "Shape_Area". The data is as follows:

	AVG_SIZE07	CROP_ACR0	AVG_SALE07	SQMI	Shape_Length	Shape_Area
	149	177626	68.29	10931	1275835.715167	16513001143.5337
	381	7609210	172.92	71297	3004175.455006	164442316568.045
	2079	18241710	94.94	14703	2859079.544323	359662292212.805
	166	529253	75.86	39555	1804888.364454	78384814218.1752
	1241	27527180	190.31	70698	1753454.614357	173072126682.401
	1401	19094311	210.8	77115	1943874.72695	187970221480.16
	2726	2576017	104.58	97813	1960001.350821	238004239410.541
	194	10116279	114.29	59822	1968992.210358	136605586997.204

The bottom of the window shows a status bar with navigation icons and the text "(0 out of 51 Selected)".

Shapefile geometry measures

- Shapefiles DO NOT create or maintain area/length fields automatically!
 - Must be created and updated manually
 - Some functions and operations can change the lengths and areas of features
 - If you find an AREA field there is no guarantee that it is correct

watersheds						
	FID	Shape *	AREA	PERIMETER	NAME	AREA_ACRE_
	0	Polygon	5265900	16620	Nameless Cave	1301.23
	1	Polygon	765900	5700	Robbinsdale Park East	189.257
	2	Polygon	1311120	27480	Red Rock Canyon	3239.8401
	3	Polygon	2283300	9780	Wonderland Drive	564.21301
	4	Polygon	511200	4140	Highway 79 Industrial Park	126.32
	5	Polygon	4786200	15780	South Robbinsdale	1182.6899
	6	Polygon	1192060	25782.6	Landfill	2945.6299

Be careful!

Table

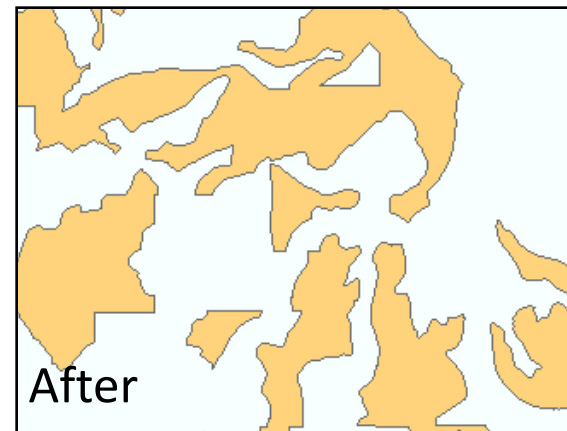
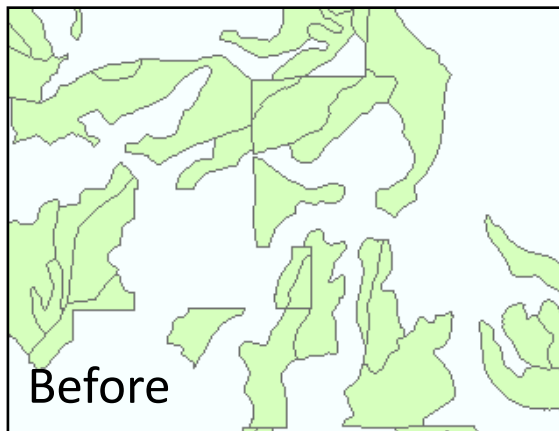
watersheds

FID	Shape *	AREA	PERIMETER	NAME	AREA_ACRE	PERCENT
0	Polygon	5265900	16620	Nameless Cave	1301.23	3.0465
1	Polygon	765900	5700	Robbinsdale Park East	189.257	0.443099
2	Polygon	1311120	27480	Red Rock Canyon	3239.8401	7.58528
3	Polygon	2283300	9780	Wonderland Drive	564.21301	1.32097
4	Polygon	511200	4140	Highway 79 Industrial Park	126.32	0.295747
5	Polygon	4786200	15780	South Robbinsdale	1182.6899	2.76898
6	Polygon	1192060	25782.6	Landfill	2945.6299	6.89646
7	Polygon	1014020	22008.6	Truck By Pass	2505.6899	5.86617

0 (0 out of 26 Selected)

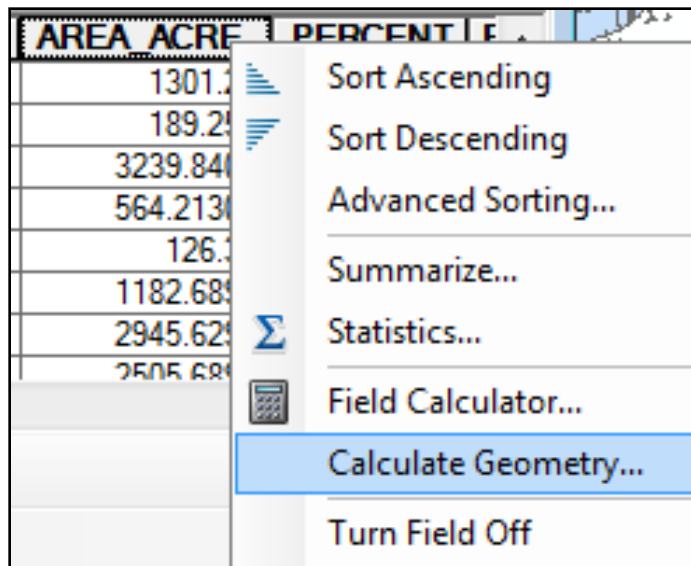
watersheds

AREA/LENGTH/PERIMETER fields in shapefiles are NOT updated when features change.

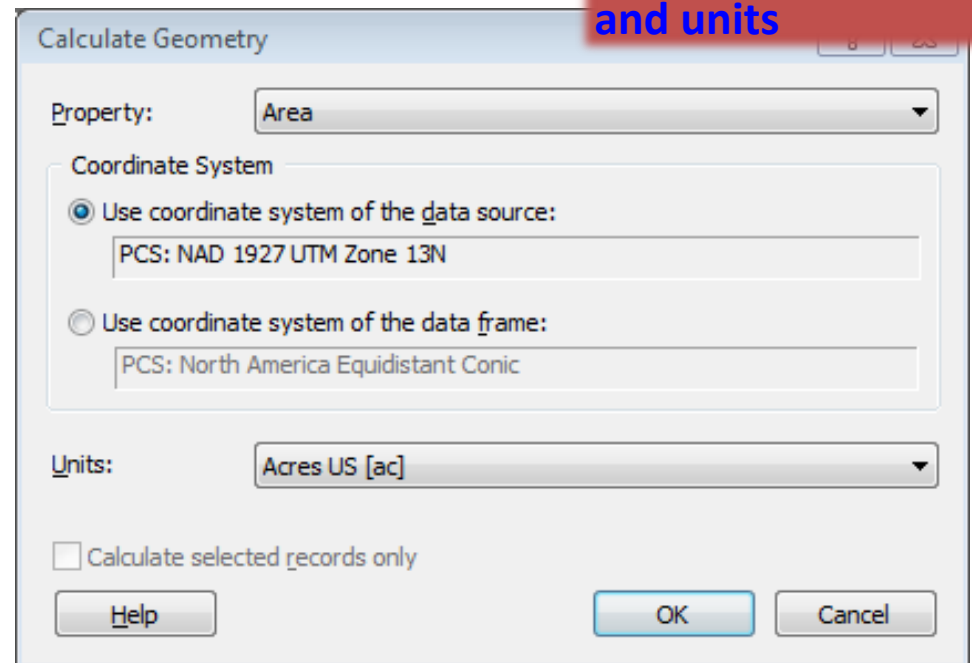


User geometry calculations

Right-click
empty or
incorrect field
to update



Choose type,
coordinate system,
and units

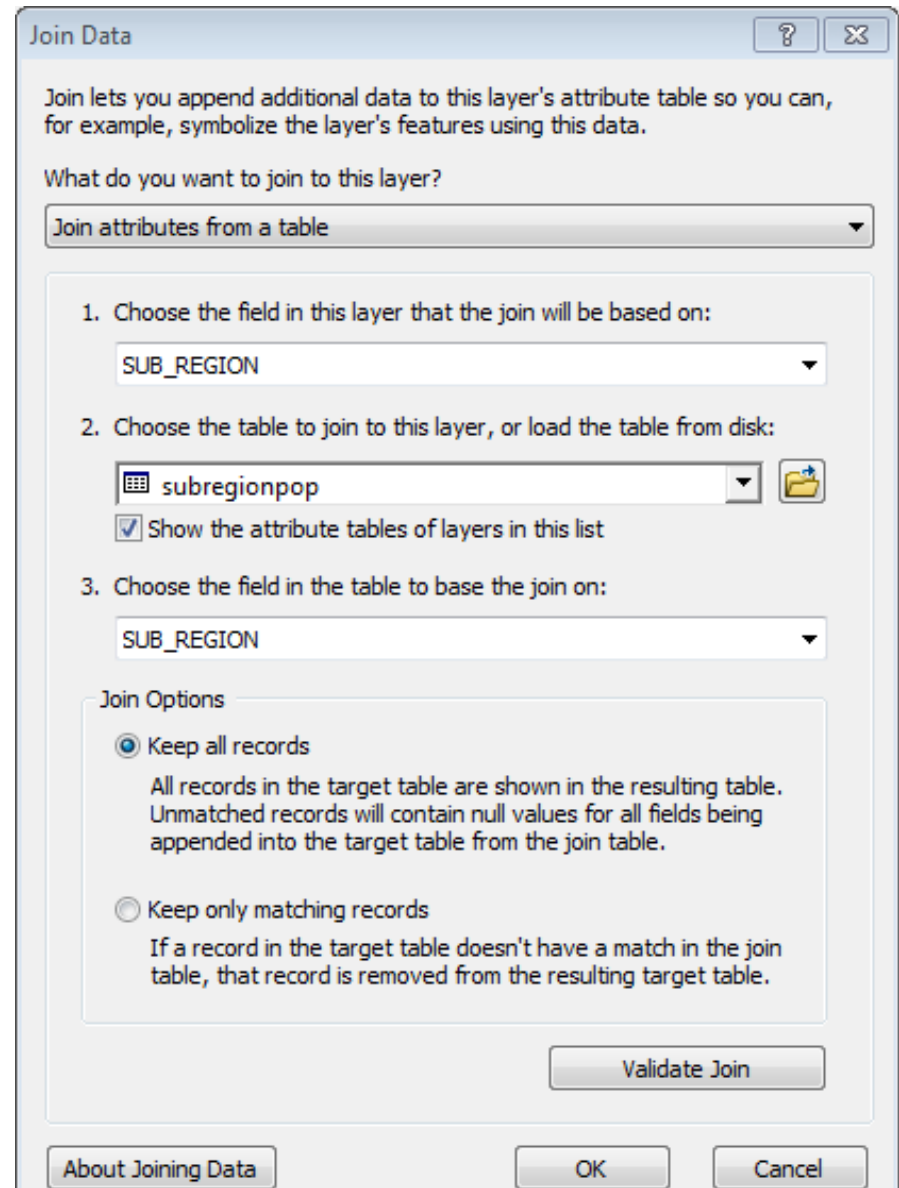
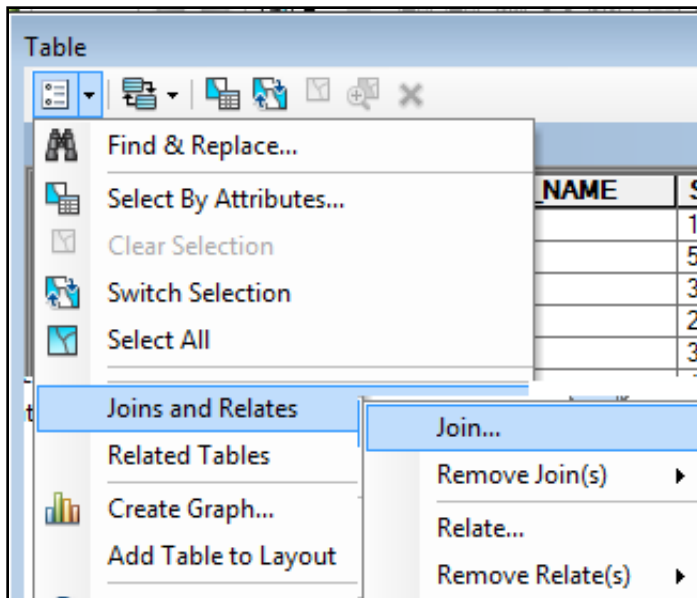
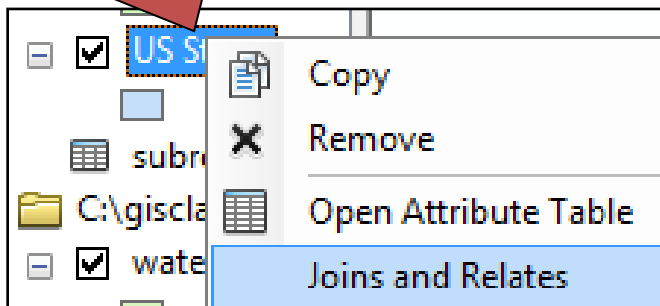


Test yourself: True or False?

- A field named AREA in a shapefile will always have the correct area?
 - **False.** Shapefile area fields are not automatically maintained.
- A field named AREA in a geodatabase will always have the correct area?
 - **False.** Automatically updated fields in a geodatabase are called Shape_Area.

How to join or relate tables

Right-click destination in the Table of Contents



Importing tables

Allowed in ArcGIS

a) Tab-delimited

b) Comma-delimited

Not allowed!

c) Fixed-column

The image displays three Notepad windows, each showing a different format for a table of data. The first window, labeled 'a', shows a tab-delimited format. The second window, labeled 'b', shows a comma-delimited format. The third window, labeled 'c', shows a fixed-column format with vertical red lines separating the columns.

a) normals_tab.txt - Notepad

NUM	STATION	JAN	FEB	MAR	APR
1	ADEL	0.89	0.97	0.93	0.93
2	ALKALI LAKE		0.52	0.46	0.79
3	ALSEA F H FALL CREEK			14.06	12.38
4	ANTELOPE 1 NW		1.64	1.27	1.26
5	APPLEGATE		3.9	3.55	2.88
6	ARLINGTON		1.39	0.99	0.76
7	ASHLAND	2.49	1.92	2.09	1.68
8	ASHWOOD 2 NE		1.67	1.3	1.22
9	ASTORIA CLATSOP CO AP			9.62	7.87
10	AUSTIN 3 S		2.72	2	2.01

b) normals_comma.csv - Notepad

NUM	STATION	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP
1	ADEL	0.89	0.97	0.93	0.93	0.83	0.66	0.38	0.3	
2	ALKALI LAKE	0.52	0.46	0.79	0.89	1.14	0.85	0		
3	ALSEA F H FALL CREEK	14.06	12.38	10.29	6.92					
4	ANTELOPE 1 NW	1.64	1.27	1.26	1.11	1.37	1.04			
5	APPLEGATE	3.9	3.55	2.88	1.95	1.18	0.64	0.2		
6	ARLINGTON	1.39	0.99	0.76	0.67	0.65	0.35	0.1		
7	ASHLAND	2.49	1.92	2.09	1.68	1.55	0.92	0.51		
8	ASHWOOD 2 NE	1.67	1.3	1.22	1.21	1.44	0.95	0		
9	ASTORIA CLATSOP CO AP	9.62	7.87	7.37	4.93	3				
10	AUSTIN 3 S	2.72	2	2.01	1.45	1.66	1.49	0.9		

c) normals_formatted.prm - Notepad

NUM	STATION	JAN	FEB
1	ADEL	0.89	0.97
2	ALKALI LAKE	0.52	0.46
3	ALSEA F H FALL CREEK	14.06	12.38
4	ANTELOPE 1 NW	1.64	1.27
5	APPLEGATE	3.9	3.55
6	ARLINGTON	1.39	0.99
7	ASHLAND	2.49	1.92
8	ASHWOOD 2 NE	1.67	1.3
9	ASTORIA CLATSOP CO AP	9.62	7.87
10	AUSTIN 3 S	2.72	2

Excel files

- Excel worksheets with suitable layout can be opened as tables in ArcGIS
- Most functions that do not involve changing the file will work (sort, query)
- Tables cannot be changed or edited

Excel file requirements

- First row must contain field headings with legal field names as defined earlier
- No blank rows or formulas should be used
- Each column should contain only text or only numbers.
- It is helpful for each column to be formatted as text, numeric, etc.
- ArcMap cannot open files that are already open in Excel—they must be closed first

Acceptable Worksheet

Columns formatted as text or numeric

Legal field names

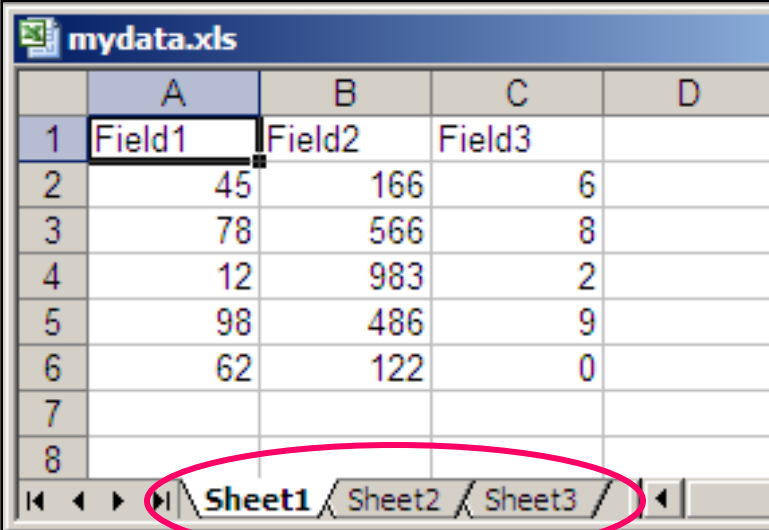
No formulas or blank lines

	D	E	F	G	H	I	J	K
1	STATION NAME	LATDEG	LATMIN	LONDEG	LONMIN	ELEV	LAT	LON
2	ADEL	42	11	119	54	4583	42.183333	-119.900000
3	ALKALI LAKE	42	58	120	0	4332	42.966667	-120.000000
4	ALSEA F H FALL CREEK	44	24	123	45	230	44.400000	-123.750000
5	ANTELOPE 1 NW	44	55	120	44	2840	44.916667	-120.733333
6	APPLEGATE	42	15	123	10	1282	42.250000	-123.166667
7	ARLINGTON	45	43	120	12	277	45.716667	-120.200000
8	ASHLAND	42	13	122	43	1724	42.216667	-122.716667
9	ASHWOOD 2 NE	44	45	120	43	2820	44.750000	-120.716667
10	ASTORIA CLATSOP CO AP	46	9	123	53	9	46.150000	-123.883333
11	AUSTIN 3 S	44	34	118	29	4213	44.566667	-118.483333
12	BAKER	44	50	117	49	3368	44.833333	-117.816667
13	BANDON 2 NNE	43	9	124	24	20	43.150000	-124.400000
14	BARNES STA	43	57	120	13	3970	43.950000	-120.216667
15		45	27	122	49	270	45.450000	-122.816667
16	B N	44	17	122	2	2152	44.283333	-122.033333
17		44	3	121	17	3660	44.050000	-121.283333

Decimal degrees = degrees + minutes/60 + seconds/3600

Workbooks and worksheets

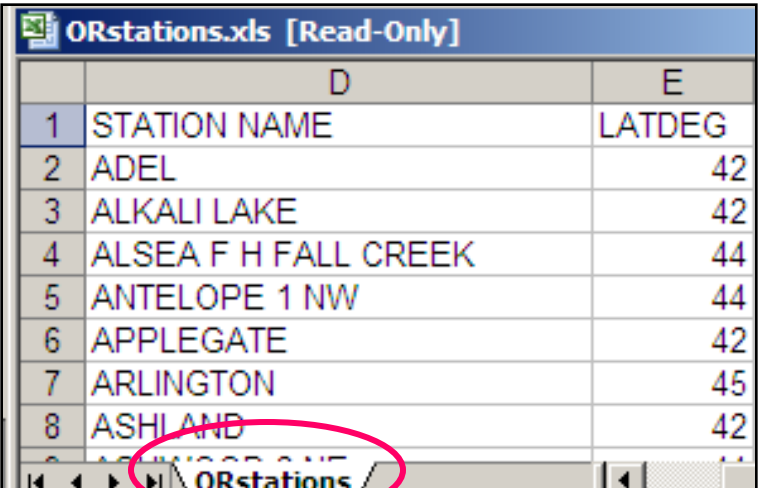
- An Excel workbook file (.xls) may contain more than one worksheet.
- By default there are three named Sheet1, Sheet2, Sheet3.
- There may be one or more named worksheets.
- ArcMap can only open one worksheet at a time.
- You will open the workbook like a folder and select a single worksheet



mydata.xls

	A	B	C	D
1	Field1	Field2	Field3	
2	45	166	6	
3	78	566	8	
4	12	983	2	
5	98	486	9	
6	62	122	0	
7				
8				

Sheet1 / Sheet2 / Sheet3



ORstations.xls [Read-Only]

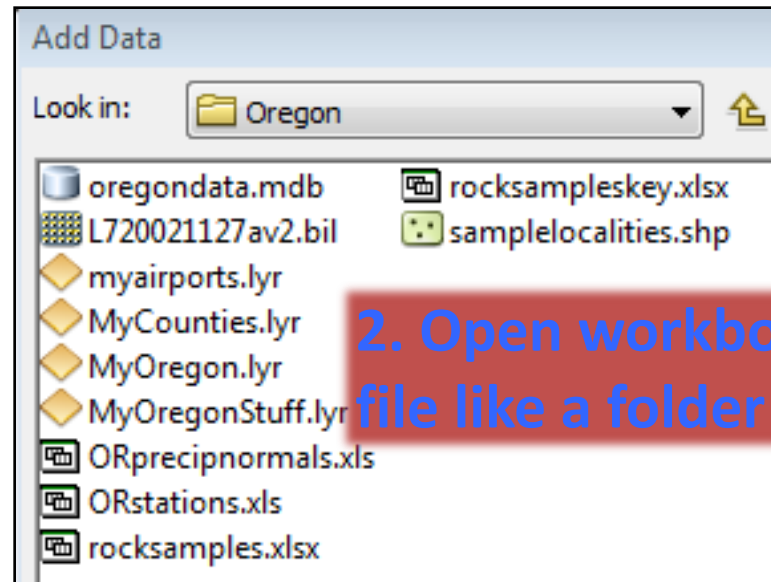
	D	E
1	STATION NAME	LATDEG
2	ADEL	42
3	ALKALI LAKE	42
4	ALSEA F H FALL CREEK	44
5	ANTELOPE 1 NW	44
6	APPLEGATE	42
7	ARLINGTON	45
8	ASHLAND	42

ORstations

Adding a worksheet

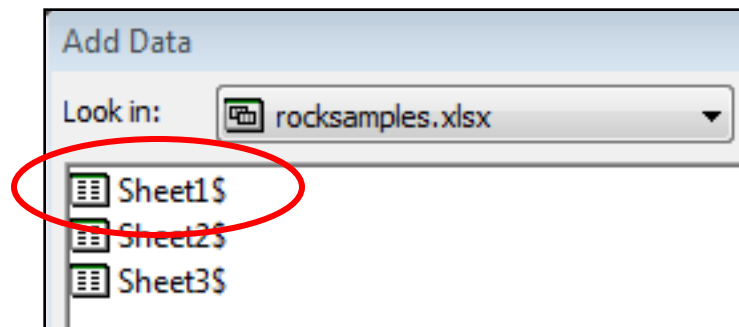
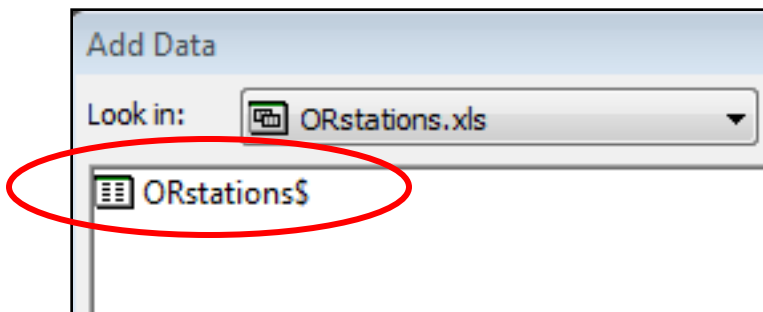


1. Add data button



2. Open workbook file like a folder

3. Add named worksheet, or the first unnamed worksheet.



Excel file in ArcMap

Table

ORstations\$

	NUM	COOPID	TYPE	LATDEG	LATMIN	LONDE	LONMIN	ELEV	LAT	LON	STATION NAME
▶	1	350036	XNP	42	11	119	54	4583	42.183333	-119.9	ADEL
	2	350118	XNP	42	58	120	0	4332	42.966667	-120	ALKALI LAKE
	3	350145	P	44	24	123	45	230	44.4	-123.75	ALSEA F H FALL CREEK
	4	350197	XNP	44	55	120	44	2840	44.916667	-120.733333	ANTELOPE 1 NW
	5	350217	P	42	15	123	10	1282	42.25	-123.166667	APPLEGATE
	6	350265	XNP	45	43	120	12	277	45.716667	-120.2	ARLINGTON
	7	350304	XNP	42	13	122	43	1724	42.216667	-122.716667	ASHLAND
	8	350312	P	44	45	120	43	2820	44.75	-120.716667	ASHWOOD 2 NE
	9	350328	XNP	46	9	123	53	9	46.15	-123.883333	ASTORIA CLATSOP CO AP
	10	350356	XNP	44	34	118	29	4213	44.566667	-118.483333	AUSTIN 3 S
	11	350412	XNP	44	50	117	49	3368	44.833333	-117.816667	BAKER
	12	350471	XNP	43	9	124	24	20	43.15	-124.4	BANDON 2 NNE
	13	350501	XNP	43	57	120	13	3970	43.95	-120.216667	BARNES STATION
	14	350595	XNP	45	27	122	49	270	45.45	-122.816667	BEAVERTON 2 SSW
	15	350652	XNP	44							BELKNAP SPRINGS 8 N
	16	350694	XNP	44							BEND

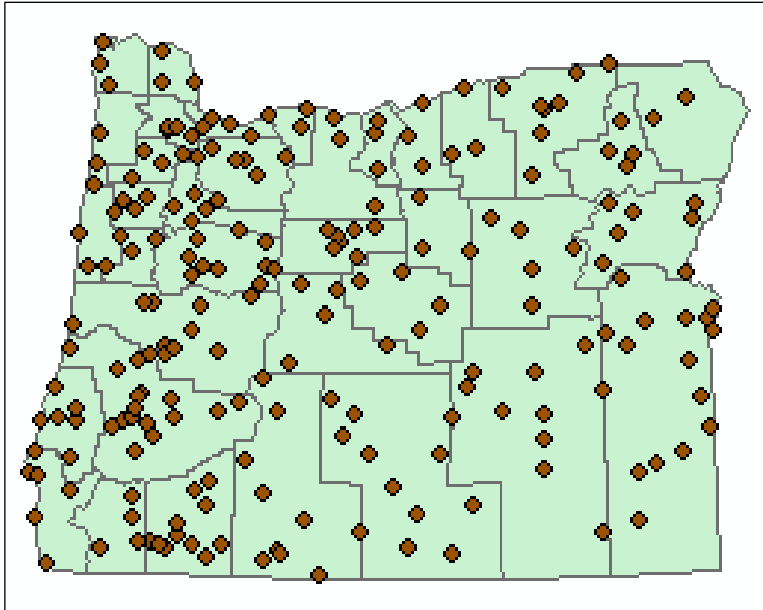
ORstations\$

(0 out of 16 rows displayed)

You can use x-y locations in a table to produce points on a map.

But first you must know the coordinate system of these locations. What is this one?

Displaying x-y values from a table

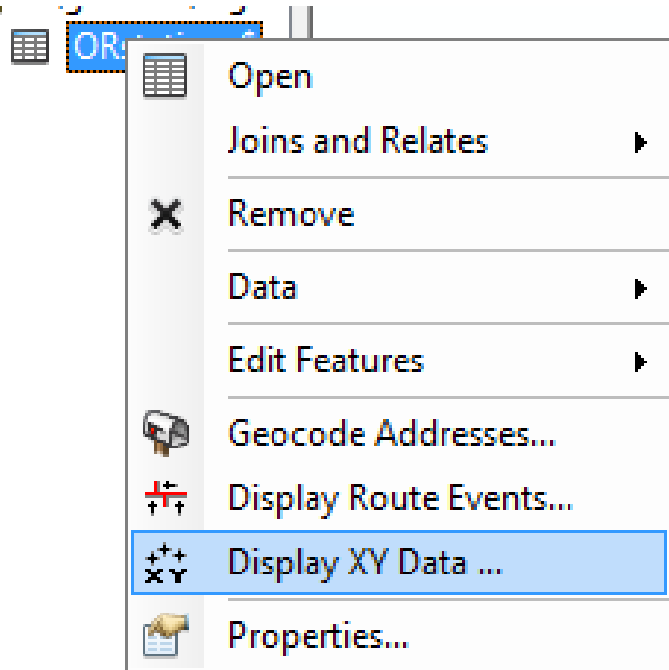


ORstations\$				
	COOPID	LAT	LON	STATION
	350036	42.183333	-119.900000	ADEL
	350118	42.966667	-120.000000	ALKALI LAKE
	350145	44.400000	-123.750000	ALSEA F H FALL
	350197	44.916667	-120.733333	ANTELOPE 1 NW
	350217	42.250000	-123.166667	APPLEGATE

Geographic coordinate system (GCS)

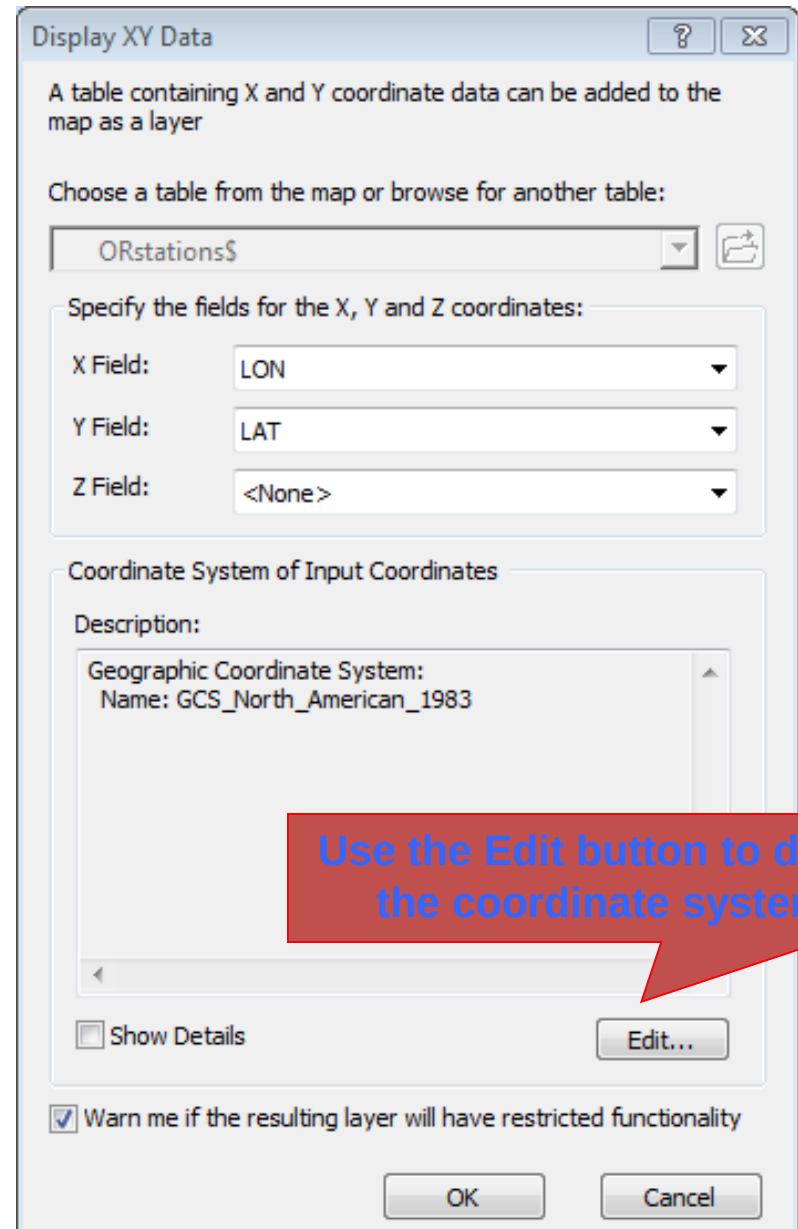
A table with x-y values can be displayed as points.
Use any coordinate system, but you must know what it is.

Display XY Data

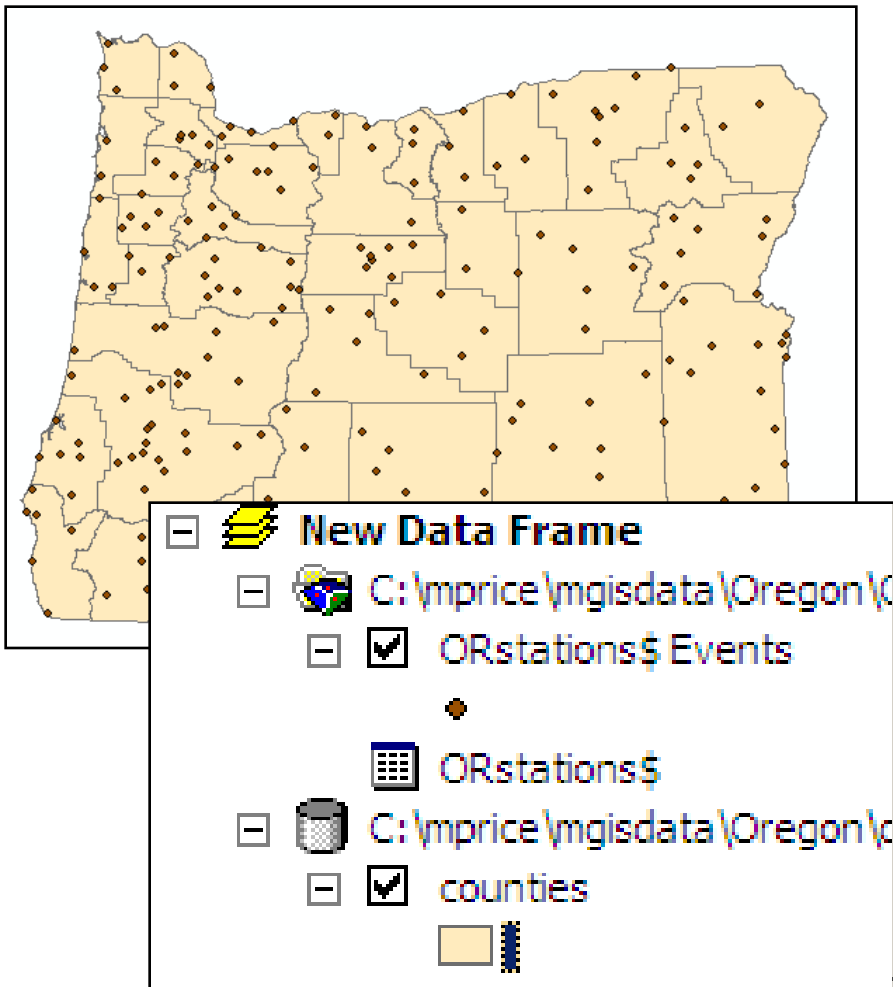


Degrees indicate a geographic coordinate system (GCS). NAD83 is the most common.

Other tables might have units in UTM meters, State Plane feet, etc. You must know the coordinate system to create correctly located points.



Oregon climate stations



- Stations appear as points on the map, an “event layer”
- Export to shapefile or feature class to keep permanently

Thank You